American Economic Review 2025, 115(8): 2689–2721
https://doi.org/10.1257/aer.20200678

Monopsony and Employer Misoptimization Explain Why Wages Bunch at Round Numbers†

By Arindrajit Dube, Alan Manning, and Suresh Naidu*

We show that administrative hourly wage data exhibit considerable bunching at round numbers. We run two experiments randomizing wages around $0.10 and $1.00 to experimentally measure left-digit bias for identical tasks on Amazon Mechanical Turk; we fail to find any evidence of discontinuity in the labor supply function at round numbers despite estimating a considerable degree of monopsony. We replicate these results in administrative worker-firm hourly wage data from Oregon. We can rule out inattention estimates found in the behavioral product market literature. We provide evidence that firms "misoptimize" wage setting. More monopsony requires less employer misoptimization to explain bunching. (*JEL* D22, J22, J31, J42)

Behavioral economics has documented a wide variety of deviations from perfect rationality among consumers and workers. For example, in the product market, prices are more frequently observed to end in $0.99 than can be explained by chance, and a sizable literature has confirmed that this reflects firms taking advantage of customers' left-digit bias (e.g Levy et al. 2011; Strulov-Shlain 2023). However, it is typically assumed that deviations from firm optimization are unlikely to survive as competition among firms drives firms that fail to maximize profits out of business. Therefore, explanations for pricing and wage anomalies typically rely on human behavioral biases. In this paper, we show that when it comes to a quantitatively important anomaly in wage setting—bunching at round numbers—it is driven by choices made by firms (or other stakeholders in wage setting) and not workers. In principle, the observed bunching can be rationalized either by "behavioral" firms forgoing significant profit in order to pay round numbers in relatively competitive labor markers, or by near-optimizing firms losing only small profits because

* Dube: University of Massachusetts Amherst, NBER, and IZA (email: adube@umass.edu); Manning: London School of Economics (email: a.manning@lse.ac.uk); Naidu: Columbia University and NBER (email: sn2430@columbia.edu). Thomas Lemieux was the coeditor for this article. We thank Sandra Baquie, Clara Berestycki, Doruk Cengiz, Len Goff, Jeff Jacobs, and Jon Zytnick for excellent research assistance. We are grateful for helpful comments from David Autor, Matt Backus, Ellora Derenoncourt, David Green, Patrick Kline, Ilyana Kuziemko, Ioana Marinescu, Jim Rebitzer, and Owen Zidar and from seminar participants at Microsoft Research, Columbia University, Boston University, Harvard University, Princeton University, IIES, the University of Toronto, the University of California, Berkeley, MIT, and the 2018 Society of Labor Economists conference. We received IRB approval for the full study from Columbia University (IRB-AAAQ9022). Experiment 1 in this paper detailed in Section III was registered in the American Economic Association's registry for randomized controlled trials under ID AEARCTR-0001349 (Dube, Manning, and Naidu 2017). Replication material can be found at Dube, Manning, and Naidu (2025).

† Go to https://doi.org/10.1257/aer.20200678 to visit the article page for additional materials and author disclosure statement(s).

employer monopsony power blunts the cost of mispricing labor. Empirically, a moderate amount of monopsony power, well within the range obtained in both our data and other recent high-quality US evidence, is sufficient to explain the substantial bunching we document in this paper with limited loss of profits.

We begin by documenting bunching in the hourly wage distribution at "round" numbers, first noted by Jones (1896). Unlike previous studies, we use data from both administrative sources and an online labor market to confirm that there is true bunching of wages at round numbers—that it is not simply an artifact of survey reporting. We begin by providing the first (to our knowledge) credible evidence on the extent to which hourly wages are bunched at round numbers in high quality, representative data on hourly wages from unemployment insurance (UI) records from the three largest US states (Minnesota, Washington, and Oregon) that collect information on hours.[1] We further assess the extent of bunching in online labor markets using a near universe of posted rewards on the online platform Amazon Mechanical Turk (MTurk).

We compare the size of the bunches in the administrative data to those in the Current Population Survey (CPS) where round number bunching is prevalent. For example, in the CPS data for 2016, a wage of \$10.00 is about 50 times more likely to be observed than either \$9.90 or \$10.10. Figure 1 shows that the hourly wage distribution from the CPS outgoing rotation group (ORG) data between 2010 and 2015 has a visually striking modal spike at \$10.00 (top panel). The middle panel of the figure shows that the share of wages ending in round numbers is remarkably stable over the past 40 years, between 30–40 percent of observations. The bottom panel of the figure also shows that between 2002 and 2016, the modal wage had been exactly \$10.00 in at least 30 states, reaching a peak of 48 in 2008. As wages have increased, we see that this number drops off in recent years, but we see a commensurate rise in the number of states with a modal wage of \$15.00. Of course, survey data is likely to overstate the degree of round number bunching. We also use a unique CPS supplement that matches respondents' wage information with that from the employers to correct for reporting error in the CPS and show that the measurement error-corrected CPS replicates the degree of bunching found in administrative data in the states where we observe both. It seems highly unlikely that such bunching at \$10.00 is present in the distribution of underlying marginal products of workers.

We first attempt to account for bunching by building and experimentally testing an imperfectly competitive model where worker behavior exhibits discontinuities at round numbers, paralleling product price studies from the marketing literature that have documented pervasive "left-digit bias" where consumers ignore lower-order digits in price. These discontinuities could result either from individual cognitive biases or social norms held by workers. What is important is that it is individual worker-side behavior exhibits the discontinuity. We design and implement two experiments on an online platform (MTurk). In the first experiment ($N = 5{,}017$), we randomly vary rewards above and below \$0.10 for the same task to estimate the labor supply function facing an online employer. Our primary sample consists

[1] We were able to use individual-level matched worker-firm data from Oregon and microaggregated data of job counts by detailed wage bins in Minnesota and Washington.

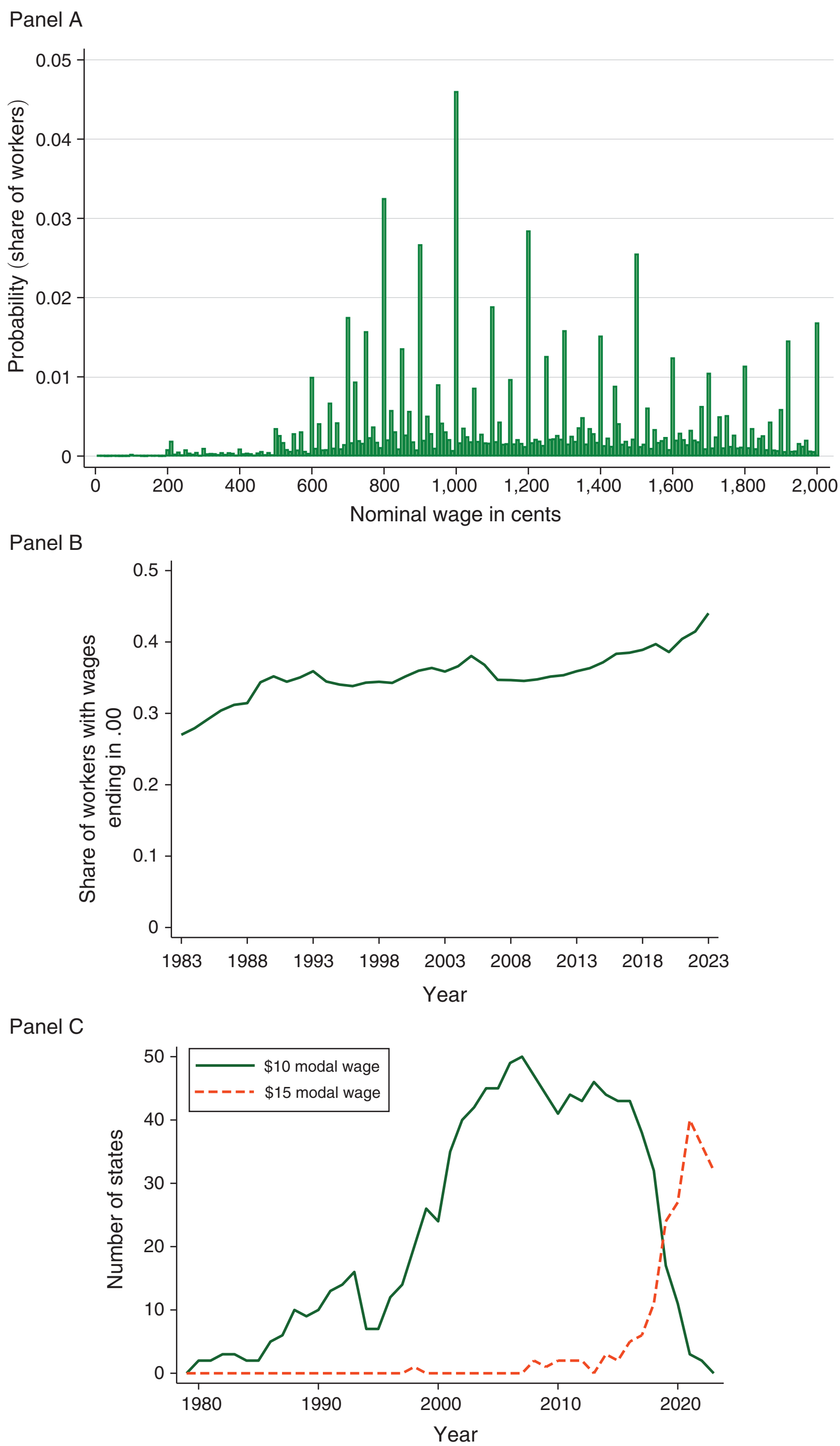

FIGURE 1. PREVALENCE OF ROUND NOMINAL WAGES IN THE CPS

Notes: The top figure shows the CPS hourly nominal wage distribution, pooled between 2010 and 2015 in $0.10 bins. The middle figure shows the fraction of hourly wages in the CPS that end in .00 from 1983 through 2023. The bottom figure shows the fraction of states with $10.00 and $15.00 modal wages in the CPS. We exclude imputed wages.

of workers who report being on the platform primarily for financial motivations, as their labor supply is more likely to respond to the rewards. Like the administrative data, the task reward distribution on MTurk exhibits considerable bunching. However, our experimentally estimated labor supply function shows no evidence of a discontinuity as would be predicted by worker left-digit bias. In a second follow-up experiment ($N = 2757$), we vary rewards around \$1.00, which also exhibits substantial bunching. Again, the estimated labor supply function exhibits no discontinuity around the \$1.00 mode. Our results are precise enough to rule out levels of left-digit bias found in the product market literature.

Next, we use administrative matched worker-firm data on hourly wages from Oregon and a matched panel design to show that there is no discontinuity in the separation elasticity at \$10.00 (the modal nominal wage during the study period) conditional on past worker wage history and other firm characteristics. These results suggest that our experimental findings on lack of left-digit bias are likely to have external validity beyond online labor markets. In addition, our analysis using the matched worker-firm data produces firm-level labor supply elasticities in the two to three range, which is consistent with estimates from other recent work (Sokolova and Sorensen 2021; Bassier, Dube, and Naidu 2022) and suggests a sizable amount of monopsony power in the US low-wage labor market.

We then extend the model to allow imperfect firm optimization in the form of employer preferences for round wages, parameterized by the fraction of profits forgone in order to pay a round number. These could be administrative costs, cognitive biases of managers, public salience, or social norms that bind on the employer rather than workers: What is important is that they constrain employer wage setting for reasons unrelated to worker behavior. We recover estimates of monopsony, employer misoptimization, and left-digit bias from the distribution of missing mass around \$10.00 relative to a smooth latent density and conclude that employer misoptimization accounts for much of the observed bunching.[2]

In the administrative data from Oregon, we further show that small businesses, employers in the construction sector (with a prevalence of contractors), and firms in industries with low adoption of frontier managerial practices are most likely to pay round numbers, consistent with bunching being a result of potentially unsophisticated employer pay setting. At the same time, we also find that bunching is the most prevalent among firms paying more labor market rent; that is, a high firm fixed effect in the sense of Abowd, Kramarz, and Margolis (1999)—hereafter, AKM. These patterns are consistent with the theoretical prediction and empirical confirmation of greater monopsony power among firms at the top of the job ladder, thereby reducing the costs of mispricing. Using our estimates of monopsony power, we calculate that employer misoptimization in US labor markets costs misoptimizing employers between 1 percent and 5 percent of profits.

Our paper is related to a growing literature on behavioral firms, which documents a number of ways firms fail to maximize profits (DellaVigna and Gentzkow 2019;

[2] While other configurations are logically possible, they do not easily explain why wages are bunched at round numbers. For example, if employers had a left-digit bias, any heaping would likely occur at \$9.99 and not at \$10.00, which is not true in reality. Similarly, if workers tended to round off wages to the nearest dollar, this would not encourage employers to set pay exactly at \$10.00. In contrast, both workers' left-digit bias and employers' tendency to round off wages provide possible explanations for a bunching at \$10.00/hour.

Goldfarb and Xiao 2011; Hortaçsu and Puller 2008; Bloom and Van Reenen 2007; Cho and Rust 2010; Romer 2006; Zwick 2021). Our paper is the first to systematically quantify the extent of employer misoptimization in the labor market, which is an old and recurring theme in institutionalist labor economics,[3] as well as the first to document the absence of left-digit bias among workers.[4] DellaVigna and Gentzkow (2019) examine two potential sources of the excess uniformity in prices they observe. The first is inertia, which includes agency as well as behavioral frictions within the firm that make it hard to set prices optimally. The second is reputational or brand concerns that may create long-term incentives to keep prices uniform despite opportunities for price discrimination. We think both of these are present in the labor market, but like Della Vigna and Gentzkow, we think the former dominates the latter in most cases. With the exception of very large public-facing brands that come under the eye of politicians and activists, most Americans have limited knowledge of what wages firms are paying, and thus public relations concerns for wage setting are likely low. Evidence that firm wage setting is influenced these extra-labor market forces comes from the recent work on national wage setting by Hazell et al. (2022). They find that many national firms set wages in a manner that is highly uniform, not taking into account local labor market conditions. Further evidence comes from the literature on voluntary minimum wages (Derenoncourt and Weil 2025), which are all at round numbers. While round number bunching is a cognitively universal bias and thus visible in many diverse labor market contexts, it is symptomatic of pervasive mispricing of labor in the face of labor market frictions. The recent literature corroborates the view that firms both have considerable leeway in wage setting and often engage in rule-of-thumb behavior, such as use of uniform and round number wages. Most closely related to this paper is Reyes (2024), who builds on our work to provide complementary evidence of bunching of wages using Brazilian data along with firm-level correlates of such bunching.[5]

In our account of wage-bunching, it is important to assume that firms have some labor market power. In this, we follow work in behavioral industrial organization that explores how firms choose prices when facing behavioral consumers in imperfectly competitive markets.[6] A recent and fast-growing literature has argued that

[3] Richard Lester, for example, noted that "Study of wage data and the actual processes of wage determination indicates that psychological, social, and historical factors . . . are highly important influences in particular cases. There appears to be a fairly high degree of irrationality in the wage structure and in company wage policies, judged either by the market analysis of economists or job evaluation within and between plants."(Lester 1946, p. 158).

[4] A large literature has discussed cognitive biases in processing price information, but little of this has discussed applications to wage determination. Behavioral labor economics has extensively documented other deviations from the standard model (e.g., time inconsistency and fairness; see Babcock et al. 2012 for an overview) for worker behavior. Worker behavioral phenomena have been replicated even in online spot labor markets (Chen and Horton 2016; DellaVigna and Pope 2018).

[5] Similar to us, Reyes (2024) finds bunching of wages is more prevalent at arguably "less sophisticated" firms, which is consistent with employer misoptimization. In this paper, we additionally provide alternative theoretical accounts of bunching (including worker left-digit bias and employer misoptimization), test these explanations including using experimental and observational evidence, and relate bunching to the presence of monopsony power in the labor market. On the other hand, Reyes (2024) extends our work by testing a hypothesis we develop below about how bunching relates to minimum wage spillovers.

[6] See Heidhues and Kőszegi (2018) for a survey and Gabaix and Laibson (2006) for an early example. Theoretical models to explain bunching in prices also assume firms have some market power: For example, Basu (1997) has a single monopolist supplying each good, Basu (2006) has oligopolistic competition, and Heidhues and Kőszegi (2008) use a Salop differentiated products model.

monopsony is pervasive in modern unregulated labor markets (Manning 2011). While some of the recent literature on monopsony has focused on identifying effects of labor market concentration on wages (Berger, Herkenhoff, and Mongey 2022; Azar, Marinescu, and Steinbaum 2022), more relevant for our paper is the literature that tries to directly estimate residual labor supply elasticities, either from firm-level shocks to productivity, rents, or wage policies (Lamadon, Mogstad, and Setzler 2022; Kline et al. 2019; Cho 2018; Derenoncourt and Weil 2025) or from direct experimental manipulation of wages (Caldwell and Oehlsen 2018; Dube et al. 2020). We show that moderate amounts of monopsony in the labor market can provide a parsimonious explanation of anomalies in the wage distribution, such as patterns of wage bunching at arbitrary numbers. To demonstrate some additional economic implications of round number bunching, we show how inertia created by round number bunching generates a new source of spillover from minimum wages and changes the interpretation of rent-sharing estimates and payroll tax incidence.

The plan of the paper is as follows. In Section I, we provide evidence on bunching at round numbers using administrative data as well as data from the CPS corrected for measurement error and benchmark these against the raw CPS results. We recover the source of the bunched observations by comparing the observed distribution to an estimated smooth latent wage distribution. We show patterns in the degree of bunching by firm characteristics. In Section II, we develop a model of bunching with worker left-digit bias and develop testable hypotheses. Section III presents findings from the online experiment, administrative data, and stated-preference experiment, recovering estimates of the degree of monopsony as well as the extent of worker left-digit bias. Section IV extends the model to account for firm misoptimization and presents estimates for the extent of misoptimization required to rationalize the observed degree of bunching with our estimated labor supply elasticities. Section V concludes.

I. Bunching of Wages at Round Numbers

There is little existing evidence on bunching of wages. One possible reason is that hourly wage data in the CPS comes from self-reported wage data where it is impossible to distinguish the rounding of wages by respondents from true bunching of wages at round numbers. Documenting the existence of wage bunching requires the use of other higher-quality data.

A. *Administrative Hourly Wage Data from Selected States*

Earnings data from administrative sources such as the Social Security Administration or UI payroll tax records are of high quality, but most do not contain information about hours. However, four states (Minnesota, Washington, Oregon, and Rhode Island) have UI systems that collect detailed information on hours, allowing the measurement of hourly wages. We have obtained individual-level matched employer-employee data from Oregon as well as microaggregated hourly wage data from Minnesota and Washington. The UI payroll records cover over 95 percent of all wage and salary civilian employment. Hourly wages are constructed by dividing quarterly earnings by total hours worked in the quarter. The

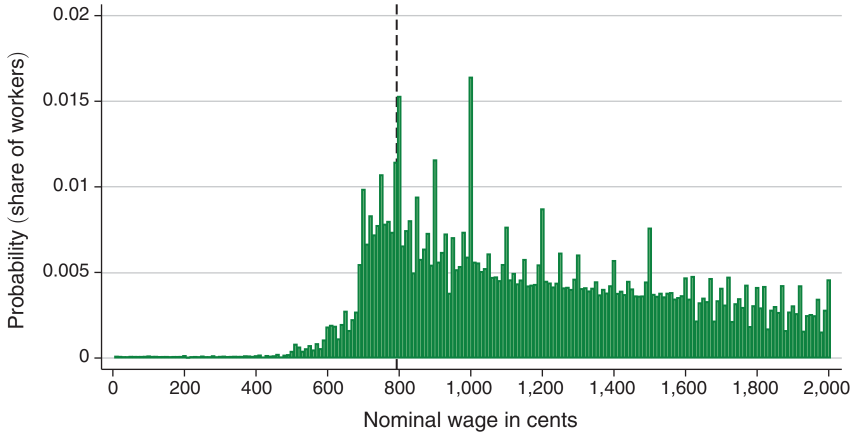

FIGURE 2. HISTOGRAM OF NOMINAL HOURLY WAGES IN POOLED ADMINISTRATIVE PAYROLL DATA FROM MINNESOTA, OREGON, AND WASHINGTON, 2003–2007

Notes: The figure shows a histogram of hourly wages in \$0.10 (nominal) wage bins, averaged over 2003:I to 2007:IV, using pooled administrative UI payroll records from the states of Oregon, Minnesota, and Washington. Hourly wages are constructed by dividing quarterly earnings by the total number of hours worked in the quarter. The counts in each bin are normalized by dividing by total employment in that state averaged over the sample period. The UI payroll records cover over 95 percent of all wage and salary civilian employment in the states. The vertical line is the highest minimum wage in the sample (7.93 in Washington in 2007). The counts here exclude NAICS 6241 and 814, home-health and household sectors, respectively, which were identified by the state data administrators as having substantial reporting errors.

microaggregated data are statewide counts of employment (and hours) by nominal \$0.05 bins between \$0.05 and \$35.00 along with a count of employment (and hours) above \$35.00. The counts exclude NAICS 6241 and 814, home health and household sectors, respectively, which were identified by the state data administrators as having substantial reporting errors.

Figure 2 shows the distribution of hourly wages in Minnesota, Oregon and Washington (we report the distributions separately in the Supplemental Appendix). The histogram reports normalized counts in \$0.10 (nominal) wage bins, averaged over 2003:I–2007:IV. We focus on this period because in later years the nominal minimum wages often reach close to \$10.00, making it difficult to reliably estimate a latent wage density in this range as we do in our analysis later in this paper. The counts in each bin are normalized by dividing by total employment. The wages are clearly bunched at round numbers, with the modal wage at the \$10.00 bin representing more than 1.5 percent of overall employment. This suggests that observed wage bunching is not solely an artifact of measurement error and is a feature of the "true" wage distribution. Further, the histogram reveals spikes at the Minnesota, Oregon, and Washington minimum wages in this period, suggesting that the hourly wage measure is accurate.

In Supplemental Appendix D, we show a very similar degree of bunching in a measurement-error corrected CPS, using the 1977 CPS Supplement that recorded wages from firms as well as workers. While the degree of bunching in the raw CPS falls with the measurement error correction, it remains substantial and indeed comparable to the administrative data.

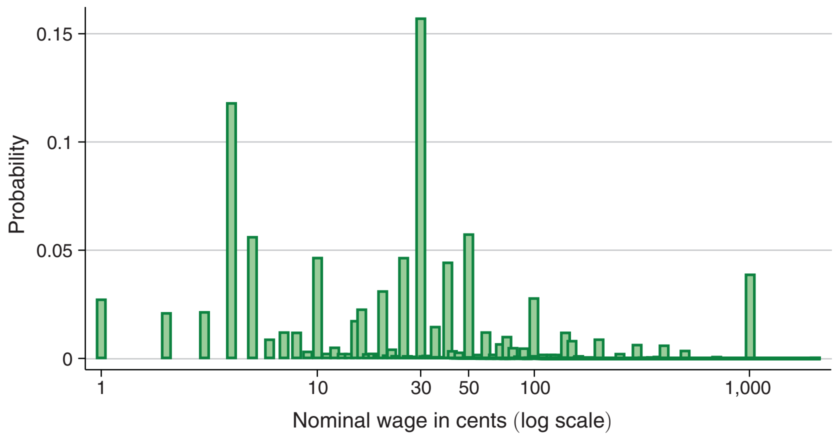

Figure 3. Histogram of Task Rewards in Online Labor Markets: MTurk

Notes: The figure shows a histogram of posted rewards by \$0.01 (nominal) bins scraped from MTurk. The *x*-axis is in log scale for ease of visualization. The sample represents all posted rewards on MTurk between May 1, 2014 and September 3, 2016.

B. *Task Rewards in an Online Market: Amazon Mechanical Turk*

Amazon MTurk is an online task market where "requesters" (employers) post small online Human Intelligence Tasks (HITs) to be completed by "Turkers" (workers).[7] Psychologists, political scientists, and economists have used MTurk to implement surveys and survey experiments (e.g., Kuziemko et al. 2015). Labor economists have used MTurk and other online labor markets to test theories of labor markets and have managed to reproduce many behavioral properties in lab experiments on MTurk (Shaw, Horton, and Chen 2011).

We obtained the universe of MTurk requesters from Panos Ipeirotis at NYU. We then used the application programming interface developed by Ipeirotis to download the near universe of HITs from MTurk from May 2014 to February 2016, resulting in a sample of over 350,000 HIT batches. We have data on reward, time allotted, description, requester ID, and first time seen and last time seen (which we use to estimate duration of the HIT request before it is taken by a worker). The data are described more fully in Supplemental Appendix F and in Dube et al. (2020).

Figure 3 shows that there is considerable bunching at round numbers in the MTurk reward distribution. The modal wage is \$0.30, with the next modes at \$0.05, \$0.50, \$0.10, \$0.40, and at \$1.00. This fact is remarkable as this is a spot labor market that has almost no regulations, suggesting the analogous bunching in real world labor markets is not driven by unobserved institutional constraints, including long-term implicit or explicit contracts. Nor is there an opportunity for bargaining, so the rewards posted are generally the rewards paid although there may be unobserved bonus payments.

[7] The subheader of MTurk is "Artificial Artificial Intelligence," and it owes its name to a nineteenth-century "automated" chess playing machine that actually contained a "Turk" person in it.

C. *Estimating Excess and Missing Mass Using a Bunching Estimator*

The excess mass in the wage distribution at a bunch that has been documented in the previous section must come from somewhere in the latent wage distribution that would result from the "nominal model" without any bunching (in the terminology of Chetty 2012).[8] This section describes how we estimate the origin of this "missing mass." We focus on the bunching at the roundest number (\$10.00 in the wage data, \$1.00 in the MTurk rewards data). We ignore the secondary bunches; this will attenuate our estimate of the extent of bunching as we will ignore the attraction that other round numbers exert on the distribution. However, we later assess robustness to controlling for secondary modes.

We assume that under the nominal model (i.e., without bunching at w_0), wages follow a latent wage distribution, $g(w)$. Following the standard approach in the bunching literature, we use a *K*th order polynomial to model the latent wage distribution $g(w)$ and estimate this excluding the area affected by bunching (i.e., the "bunching interval") that ranges between $w_0 - \Delta w$ and $w_0 + \Delta w$ around the bunching point, w_0. Within the bunching interval, we model the bunching effects using \$0.10 bins, indicated by j going up by \$0.10 increments from $w_0 - \Delta$ to $w_0 + \Delta$.[9] (See Kleven 2016 for a discussion of this type of strategy used in the literature.)

We use bin-level counts of wages c_w in \$0.10 bins and define $p_w = \frac{c_w}{\sum_{j=0}^{\infty} c_j}$ as the normalized count or probability mass for each bin. We then estimate

$$p_w = \sum_{j=w_0-\Delta w}^{w_0+\Delta w} \beta_j \mathbf{1}_{\{w=j\}} + \sum_{i=0}^{K} \alpha_i w^i + \epsilon_w. \tag{1}$$

In this expression, j sums over \$0.10 wage bins within the bunching interval, and $\sum_{i=0}^{K} \alpha_i w^i$ forms the *K*th order polynomial approximating $g(w)$, while the β_j terms are coefficients on dummies for bins in the excluded range around w_0, between $w_L = w_0 - \Delta w$ and $w_H = w_0 + \Delta w$. β_{w_0} is the excess bunching (*EB*) at w_0. In addition, $\sum_{j=w_0-\Delta w}^{w_0-10} \beta_j$ is the missing mass strictly below w_0 (*MMB*), while $\sum_{j=w_0+10}^{w_0+\Delta w} \beta_j$ is the missing mass strictly above w_0 (*MMA*). However, it might be likely that the number of workers is increasing in w, so our test for left-digit bias is whether $\sum_{j=w_0-\Delta w}^{w_0-10} \frac{\beta_j}{g(j)} = \sum_{j=w_0+10}^{w_0+\Delta w} \frac{\beta_j}{g(j)}$. In other words, we test whether the share of workers who would have been paid the latent wage j but are instead paid w_0 is

[8] Following the literature, our procedure assumes that the missing mass is originating entirely from the surrounding bunching interval. In principle, it is possible that the missing mass is originating from latent nonemployment—that is, jobs that would not exist under the nominal model in the absence of bunching. However, given the extent to which some of the excess jobs at \$10 are coming from latent nonemployment, one would need to assume either that (i) these jobs have latent productivity exactly at \$10.00 so that employers are indifferent between entering and not entering or that (ii) they have productivity greater than \$10 but have a fixed cost of not paying exactly \$10 that is independent of the size of the profits from paying different wages under the nominal model. Both of these assumptions strike us as implausible. As an empirical matter, if some of the excess mass at \$10 is originating from latent nonemployment, the estimated missing mass around \$10 would be smaller in magnitude than the excess mass at \$10. However, our estimated missing mass from the surrounding interval is indeed able to account for the size of the excess mass, which suggests that latent nonemployment is unlikely to be an important contributor to the excess mass in our case.

[9] We use \$0.01 bins in the MTurk data.

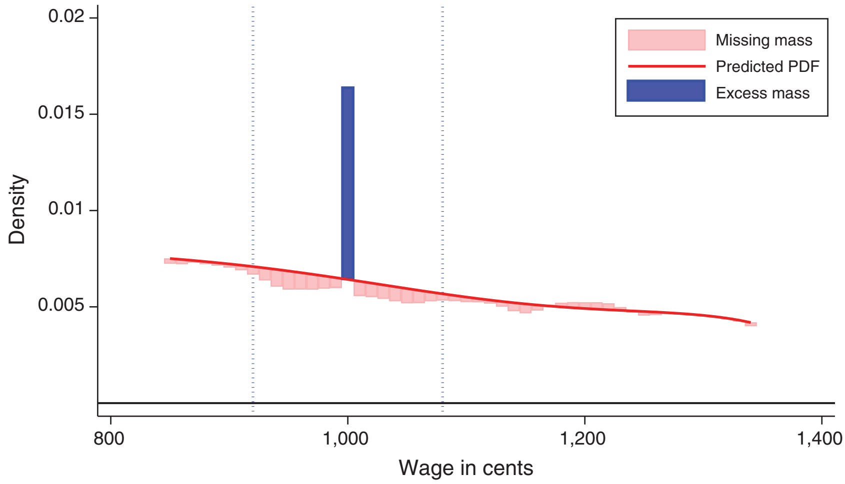

FIGURE 4. EXCESS BUNCHING AND MISSING MASS AROUND \$10.00 USING ADMINISTRATIVE DATA ON HOURLY WAGES (MINNESOTA, OREGON, WASHINGTON)

Notes: The reported estimates of excess bunching at \$10.00 and missing mass in the interval around \$10.00 as compared to the smoothed predicted probability density function, using administrative hourly wage counts from Oregon, Minnesota, and Washington, aggregated by \$0.10 bins, over the 2003:I–2007:IV period. The darker shaded blue bar at \$10.00 represents the excess mass, while the lighter red shaded region represents the missing mass. The dotted lines represent the estimated interval from which the missing mass is drawn. The predicted PDF is estimated using a sixth-order polynomial, with dummies for each \$0.10 bin in the interval from which the missing mass is drawn. The width of the interval is chosen by iteratively expanding the interval until the missing and excess masses are equal, as described in the text.

different when considering bunchers from above versus below (averaged over all j within the bunching interval).

Since Δw is unknown, we use an iterative procedure similar to Kleven and Waseem (2013). Starting with $\Delta w = 10$, we estimate equation (1) and calculate the excess bunching EB and compare it with the missing mass $MM = MMA + MMB$. If the missing mass is smaller in magnitude than the excess mass, we increase Δw and reestimate equation (1). We do this until we find a Δw such that the excess and missing masses are equalized. Since Δw is itself estimated, we estimate its standard error using a bootstrapping procedure suggested by Chetty (2012) and Kleven (2016). In particular, we resample (with replacement) the errors $\hat{\epsilon}_w$ from equation (1) and add these back to the fitted $\hat{p}_w$ to form a new distribution $\tilde{p}_w$, and estimate regression (1) using this new outcome. We repeat this 500 times to derive the standard error for Δw. The estimate of Δw and its standard error will be useful later for the estimation of other parameters of interest.

In Figure 4 we show the estimates for the administrative data from Minnesota, Oregon, and Washington using polynomial order $K = 6$. For visual ease, we plot the kernel-smoothed $\hat{\beta}_j$ for the missing mass. Even leaving out the prominent spike at \$10.00, the wage distribution is not smooth and has relatively more mass at multiples of \$0.05, \$0.10 and \$0.25. For this reason, it is easier to detect the shape of the missing mass by looking at the kernel-smoothed $\hat{\beta}_j$. Moreover, we show the excess and missing mass relative to the counterfactual $\widehat{p_w^C} = \sum_{i=0}^{6} \alpha_i w^i$. There is clear bunching at \$10.00 in the administrative data, consistent with evidence from

TABLE 1—ESTIMATES FOR EXCESS BUNCHING, MISSING MASS, AND INTERVAL AROUND THRESHOLD

	(1)	(2)	(3)	(4)
Value of w_0	\$10.00	\$10.00	\$10.00	\$10.00
Excess mass at w_0	0.010	0.034	0.014	0.041
	(0.002)	(0.008)	(0.004)	(0.009)
Total missing mass	−0.013	−0.043	−0.016	−0.033
	(0.008)	(0.033)	(0.012)	(0.045)
Missing mass below	−0.007	−0.024	−0.009	−0.019
	(0.007)	(0.020)	(0.009)	(0.027)
Missing mass above	−0.006	−0.019	−0.007	−0.014
	(0.008)	(0.028)	(0.010)	(0.036)
Test of equality of missing shares of latent $<$ and $>$ 0:				
t-statistic	−0.338	−0.097	−0.065	−0.151
Bunching $= \frac{Actual\ mass}{Latent\ density}$	2.555	6.547	4.172	8.394
	(0.353)	(6.425)	(1.742)	(7.557)
w_L	\$9.20	\$9.30	\$9.30	\$9.30
w_H	\$10.80	\$10.70	\$10.70	\$10.70
$\omega = \frac{(w_H - w_0)}{w_0}$	0.080	0.070	0.070	0.070
	(0.026)	(0.033)	(0.034)	(0.033)
Data:	Admin OR and MN and WA	CPS-Raw OR and MN and WA	CPS-MEC OR and MN and WA	CPS-Raw

Notes: The table reports estimates of excess bunching at threshold w_0, missing mass in the interval around w_0 as compared to the smoothed predicted probability density function, and the interval (ω_L, ω_H) from which the missing mass is drawn. It also reports the t-statistic for the null hypothesis that the size of the missing mass to the left of w_0 is equal to the size of the missing mass to the right. The predicted PDF is estimated using a sixth-order polynomial, with dummies for each bin in the interval from which the missing mass is drawn. The width of the interval is chosen by iteratively expanding the interval until the missing and excess masses are equal, as described in the text. In columns 1–3, estimates are shown for bunching at \$10.00 from pooled Minnesota, Oregon, and Washington using the administrative hourly wage counts, the raw CPS data, and measurement error corrected CPS (CPS-MEC) over the 2003:I–2007:IV period. Bootstrap standard errors based on 500 draws are in parentheses.

the histogram above. We find that the excess bunching can be accounted for by missing mass spanning $\Delta w = \$0.80$; we can also divide Δw by w_0 and normalize the radius as $\omega = (w_H - w_0)/w_0 = 0.08$. Visually, the missing mass is coming from both below and above \$10.00, which will be relevant when we consider alternative explanations for bunching.

These estimates are also reported in Table 1, column 1. The bunch at \$10.00 is statistically significant with a coefficient of 0.010 and standard error of 0.002. In addition, the size of the missing mass from above and below w_0 are quantitatively very close at −0.006 and −0.007 respectively. The t-statistic for the null hypothesis that the missing mass (relative to latent) is equal for bunchers from above and below is 0.338. This provides strong evidence against worker left-digit bias, which would have implied an asymmetry in the missing masses. The radius of the missing mass interval is $\omega = 0.08$ with a standard error of 0.026. In other words, employers who are bunching appear to be paying as much as 8 percent above or below the wage that maximizes profits under the nominal model.

In column 2, we use the CPS data limited to Minnesota, Oregon, and Washington only. We find a substantially larger estimate for the excess mass, around 0.041, consistent with rounding of wages by survey respondents. In column 3, we report estimates using the reweighted CPS counts for Minnesota, Oregon, and Washington adjusted for

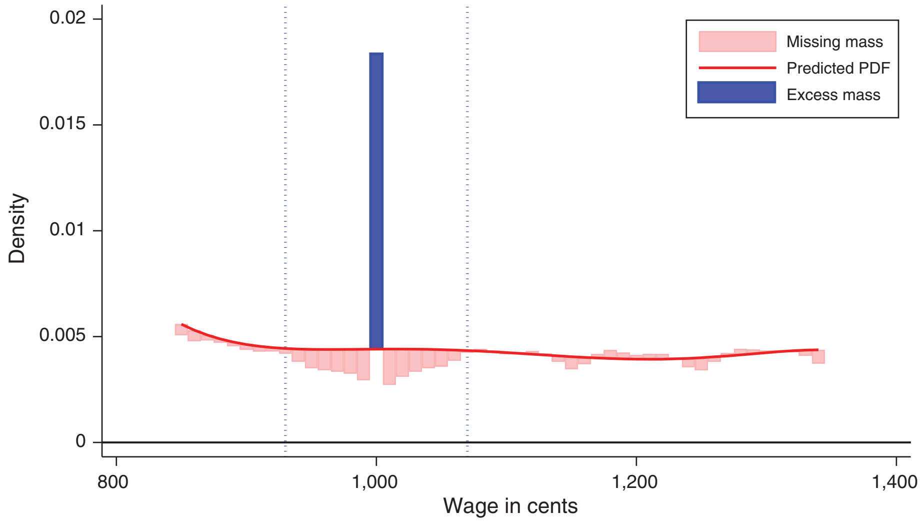

Figure 5. Excess Bunching and Missing Mass around $10.00 Using Measurement Error Corrected CPS Data

Notes: The reported estimates of excess bunching at $10.00 and missing mass in the interval around $10.00 as compared to the smoothed predicted probability density function, using CPS data for Minnesota, Oregon, and Washington, corrected for measurement error using the 1977 administrative supplement. The darker shaded blue bar at $10.00 represents the excess mass, while the lighter red shaded region represents the missing mass. The dotted lines represent the estimated interval from which the missing mass is drawn. The predicted PDF is estimated using a sixth-order polynomial, with dummies for each $0.10 bin in the interval from which the missing mass is drawn. The width of the interval is chosen by iteratively expanding the interval until the missing and excess masses are equal, as described in the text.

rounding due to reporting error using the 1977 supplement (we refer to this measurement error corrected estimate as CPS-MEC). The CPS estimate of bunching adjusted for measurement error is much closer to the administrative data, with an estimated magnitude of 0.014; while it is still somewhat larger, we note that the estimate from the administrative data is within the 95 percent confidence interval of the CPS-MEC estimate. The use of the CPS supplement substantially reduces the discrepancy, which is reassuring. At the same time, we note that the estimates for ω using the CPS (0.07) are remarkably close to those using the administrative data (0.08). The graphical analogue of column 3 is in Figure 5.

In Supplemental Appendix B we probe the robustness of these bunching estimates to a variety of specifications of the latent distribution. We include indicators for $0.25 and $0.50 modes, higher degree polynomials, and Fourier bases. We also estimate a specification that uses a polynomial in the real (rather than the nominal) wage distribution as the latent distribution. This is a "difference-in-bunching" specification that exploits the fact that the nominal bunching point maps to different locations in the underlying real wage distribution across periods. This divergence allows us to observe the latent real wage distribution in one period without interference from bunching in the other. In doing so, we avoid relying on strong parametric assumptions about the shape of the nominal wage distribution—a drawback of standard bunching estimators highlighted by Blomquist et al. (2021). The radius of the missing mass and ratio of the excess to missing mass are quite similar across these specifications.

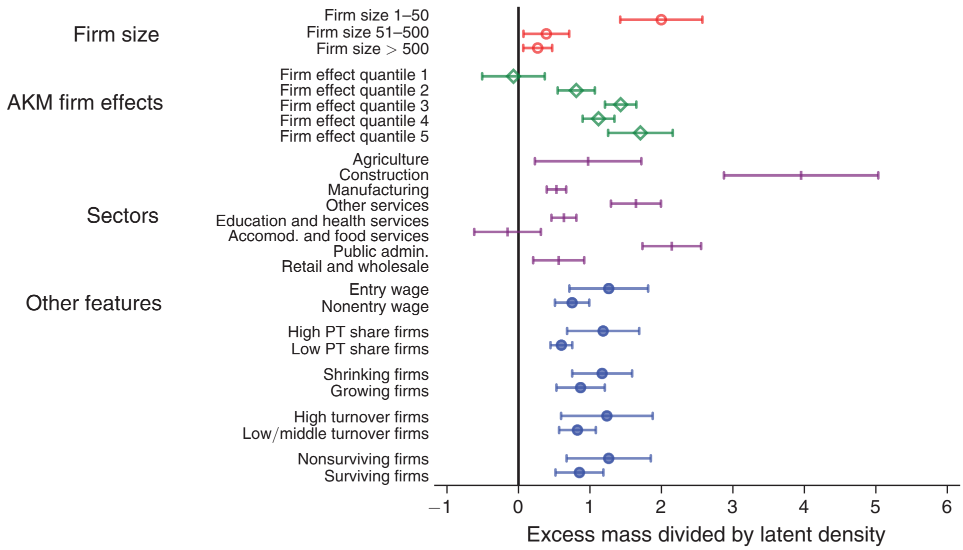

Figure 6. Heterogeneity in Bunching by Oregon Firm Characteristics

Notes: The figure plots the extent of bunching (excess mass at \$10 divided by latent density at that wage) for different subgroups. An estimate greater than zero indicates bunching. AKM firm effects refer to deciles of firm effects estimated via Abowd, Kramarz, and Margolis (1999). PT refers to part-time workers (less than 20 hours); high PT share firms are those in the top quartile. High turnover firms have separation rates in the top quartile. Shrinking firms saw employment fall over the sample (2003–2007), while growing firms saw employment weakly increase. Nonsurviving firms are those that were no longer in the dataset by 2007:IV. Ninety-five percent confidence intervals are based on standard errors clustered at the wage bin level.

The main conclusions from this section are that the missing mass seems to be drawn symmetrically from around the bunch and from quite a broad range. As the model below shows, these facts are informative about possible explanations for bunching and the nature of labor markets.

D. *Which Employers Pay Round Numbers?*

In his original paper, Jones (1896) explained the use of round numbers in wages as a result of difficulties and uncertainties in measuring productivity, the costs of calculating and administering wages, and the influence of social norms.[10] These factors are considered the costs of setting nonround wages for firms.

We begin by analyzing which types of firms are more likely to bunch wages exactly at \$10.00. If bunching is driven by managerial cognition or employer administrative costs, then firms that bunch should exhibit characteristics suggestive of "behavioral" factors, such as a lack of administrative sophistication or modern management practices. We use various firm characteristics from the matched employer-employee data from Oregon. Figure 6 presents the extent of bunching (ratio of excess mass to latent wage density) for different groups of firms, categorized by firm size, quintiles of

[10] As Jones (1896, p. 114) noted, "Round numbers frequently locate for us those fields in which propriety dictates that economic motives, which prompt exact measurements of value, shall be made subordinate to other considerations."

hourly wage firm effects,[11] industry, the share of part-time workers (working less than 20 hours), and for starting wages.

The most significant pattern observed is that bunching is pervasive across firms of various industries, sizes, and wage policies. In all but one case, the extent of bunching is statistically significant at least at the 90 percent confidence level and typically at the 95 percent level. The sole exception is the accommodation and food services sector, where Oregon's minimum wage (indexed to inflation) is highly binding.

At the same time, there are meaningful differences in bunching that are worth noting. We find that firms most likely to bunch are very small firms, particularly in the construction sector, which includes many small contractors. This is consistent with these firms being relatively "unsophisticated," more likely to pay in cash, and less likely to have standardized pay practices (e.g., automatic inflation escalation) that would remove employer discretion in wage setting. As a result, these firms are more prone to bunching wages at round numbers. The public sector also shows unusual levels of bunching, which could reflect low cost-minimization pressures in public wage setting, or it may indicate a preference among public sector workers or unions for benefits over wages. Entry-level wages are also more likely to bunch, which aligns with the idea that firms may use other mechanisms for wage increases (e.g., applying a uniform percentage increase to all workers), which gradually erode the tendency to bunch wages at round numbers over time. Importantly, and consistent with Reyes (2024), we find that bunching is more prevalent among firms that exited prior to the end of the sample period, among firms with high turnover, and to an extent among firms that shrank in employment over the sample period. Overall, the variation in round number wage use across firms supports the hypothesis that bunching is more prevalent among less sophisticated firms.

Although we lack extensive firm-level data on human resource practices to merge with the Oregon data, we incorporate the World Management Survey database from Bloom and Van Reenen (2007), aggregated to the industry level. Supplemental Appendix Figure A.3 shows the prevalence of bunching by the degree of adoption of advanced human resource management practices. Overall, the matched subsample exhibits less bunching as the management scores are available only for the manufacturing sector where bunching is less common. However, within this sample the extent of bunching decreases with managerial sophistication, with the lowest tercile showing the greatest bunching. This further supports the idea that less sophisticated firms are more likely to bunch.

Finally, we find that the firms with higher AKM effects are more likely to bunch; we return to this point later in the paper in discussing how monopsony power may affect bunching.

II. A Model of Round Number Bunching in the Labor Market

As a first pass for explaining the bunching at round numbers, this section presents a model of the labor market where workers exhibit a left-digit bias, mirroring inattention in the product market. Suppose there is a mass one of workers differing

[11] These firm effects are estimated using the Abowd, Kramarz, and Margolis (1999) decomposition. The Oregon-specific hourly wage decomposition results are detailed in Bassier, Dube, and Naidu (2022).

in their marginal product p, assumed to have density $k(p)$ and CDF $K(p)$—assume labor is supplied inelastically to the market as a whole. We assume there is only one "round number" wage in the vicinity of the part of the productivity distribution we consider—denote this by w_0.

We model the left-digit bias of workers following the recent literature on left-digit bias in product markets. Starting from a discrete choice model of the labor market, a workers i's utility at a job j is

$$u_{ij} = \eta_1\big((1-\theta)\,w_i + \theta\lfloor w_i\rfloor\big) + \epsilon_{ij}. \tag{2}$$

Here ϵ_{ij} captures idiosyncratic valuations of the job that are independently drawn from a Type I extreme value distribution. Importantly, here there can be a discontinuous jump in the utility at a round number $w_0 = \lfloor w_j\rfloor$ due to left-digit bias among workers (where $\lfloor\cdot\rfloor$ is the "floor" or "greatest integer" function). Left-digit bias has been documented in a wide variety of markets, used to explain prevalence of product prices that end in 9 or 99, and is a natural candidate explanation for bunching in the wage distribution.[12]

This model implies a logit labor supply function that is linear in wage with a jump. When the number of jobs J is large, we get a log-linear labor supply curve given by

$$\ln(l_j) = \underbrace{\eta_1(1-\theta)}_{\eta_0}w_j + \underbrace{\eta_1\theta}_{\gamma_0}\lfloor w_j\rfloor. \tag{3}$$

We estimate variants of this specification in our empirical work below and recover an estimate of $\theta = \gamma_0/(\gamma_0+\eta_0)$. For our theoretical results, in order to obtain tractable closed-form formulas for bunching, we will work with a constant elasticity specification that takes the standard log-log labor supply specification and augments it with a wage-floor term. We also implement this specification empirically and find very similar estimates of θ as the linear specification. We thus report results from both the linear specification above and log specification, given below as

$$\ln(l_j) = \eta\ln(w_j) + \gamma_1\lfloor w_j\rfloor. \tag{4}$$

When a sample is restricted to the vicinity of a specific round number $w_o = \lfloor w_j\rfloor$, we can recover our prespecified regression where $\gamma = \gamma_1 \times w_0$:

$$\ln(l_j) = \eta\ln(w_j) + \gamma\cdot\mathbf{1}_{\{w_j\geq w_0\}}. \tag{5}$$

[12]For example, Levy et al. (2011) show that 65 percent of prices in their sample of supermarket prices end in nine (33.4 percent of internet prices), and prices ending in nine are 24 percent less likely to change than prices ending in other numbers. Snir et al. (2012) also document asymmetries in price increases versus price decreases in supermarket scanner data, consistent with consumer left-digit bias. A number of field and lab experiments document that randomizing prices ending in nine results in higher product demand (Anderson and Simester 2003; Thomas and Morwitz 2005; Manning and Sprott 2009; Pope, Pope, and Sydnor 2015). Pope, Pope, and Sydnor (2015) show that final negotiated housing prices exhibit significant bunching at numbers divisible by $50,000, suggesting that round number focal points can matter even in high stakes environments. Lacetera, Pope, and Sydnor (2012) show that car prices discontinuously fall when odometers go through round numbers such as 10,000. Allen et al. (2016) document bunching at round numbers in marathon times and interpret this as reference-dependent utility. Backus, Blake, and Tadelis (2019) show that posted prices ending in round numbers on eBay are also a signal of willingness to bargain down.

It can be readily seen that $\theta \approx \gamma_1/(\gamma_1 + \eta)$ with the γ and η from equation (5).[13]

When worker productivities p vary across firms, wages are dispersed and depend on p. Define $\rho(w,p) = (p - w)\, l^*(w,p)$. Here $\rho(w,p)$ is, in the language of Chetty (2012), the "nominal model" that parameterizes profits in the absence of left-digit bias. Optimizing wages in the nominal model would yield a smooth "primitive" profit function of productivity given by $\pi(p_j) = (p_j/1 + \eta)^{1+\eta}$, but the presence of worker biases induces discontinuities in true profits at round numbers. In deciding on the optimal wage for employers, one simply needs to compare the profits to be made by maximizing the nominal model and paying the round number. Consider the wage that maximizes the nominal model. Given the isoelastic form of the labor supply curve to the individual firm, this can simply be shown to be

$$w^*(p) = \left(\frac{\eta}{1 + \eta}\right)p, \tag{6}$$

where the size of the markdown on the marginal product is determined by the extent of competition in the labor market. If the labor market is perfectly competitive, $\eta \to \infty$, and wages are equal to the marginal product. We will refer to the wage that maximizes the nominal model as the latent wage. The firm with job productivity p will pay the round number wage as opposed to the latent wage if

$$e^{\gamma \mathbf{1}_{\{w^*(p) < w_0\}}} > \frac{\rho(w^*(p),p)}{\rho(w_0,p)}. \tag{7}$$

Taking logs, we obtain that a firm will pay the round number if $p \in \left[p_*, \frac{(1 + \eta)}{\eta} w_0\right]$, where p_* solves

$$\gamma = \ln \rho(w^*(p_*),p_*) - \ln \rho(w_0,p_*). \tag{8}$$

This implies that all jobs with latent wages between w_0 and $w^*(p_*)$ will pay w_0, and bunching will be larger the greater is γ. For the marginal bunching job, the proportionate increase in profits from paying a round number wage is γ. One immediate implication of this result is that the patterns of excess and missing mass documented in Section IC are not consistent with left-digit bias. That is, even without direct estimates of γ and η, the fact that the excess mass at 10.00 is accounted for by equal amounts of missing mass from both above and below 10.00 suggests that left-digit bias is not the immediate determinant of bunching at round numbers. We now turn toward directly identifying estimates of η and γ, and hence θ, from experimental and observational data.

[13] Note that for w_j close to w_0 we have the elasticity at the round number w_0 given by $w_0 \frac{d\log(l_j)}{dw_j} = \left.\frac{d(\eta_1 \log((1 - \theta)w_j + \theta\lfloor w_j \rfloor))}{dw_j}\right|_{w_j = w_0} = \eta_1$, which would be given by $\eta + \gamma$ estimated from equation (5), and the elasticity below the round number given by $w_j \left.\frac{d(\eta_1 \log((1 - \theta)w_j + \theta\lfloor w_j \rfloor))}{dw_j}\right|_{w_j \neq w_0} = (1 - \theta)\eta_1$, which would be estimated as η from equation (5). So θ and η_1 can be recovered from η and γ (and w_0).

III. Experimental and Observational Evidence on Worker Left-Digit Bias

A. *Experimental Evidence from MTurk*

While the observational and survey experiment evidence strongly points away from worker left-digit bias, one might still worry about omitted variables bias or survey responses not reflecting real stakes. Further, neither of the earlier experiments allow us to test whether productivity also responds to the wage. We therefore supplement these approaches by conducting two high-powered experiments on MTurk to test for left-digit bias. Online labor market experiments have a high degree of internal validity though a disadvantage is that one is inevitably unsure about the external validity of the estimates. For example, one might expect that these "gig economy" labor markets are very competitive because they are lightly regulated and there are large numbers of workers and employers with few informational frictions or long-term contracting. However, in a companion paper Dube et al. (2020) compile labor supply elasticities implicit in the results from a number of crowdsourcing compensation experiments on MTurk, finding a precision-weighted elasticity of 0.14 across five experiments (including the first one in this paper). This low estimate implies considerable market power in these types of crowdsourcing labor markets, widely used to obtain training data for machine learning applications (Kingsley, Gray, and Suri 2015).

In their original paper on labor economics on MTurk, Horton, Rand, and Zeckhauser (2011) implement a variant of the experiments we conduct below, making take-it-or-leave-it offers to workers with random wages in order to trace out the labor supply curve. However, while they label this an estimate of labor supply to the *market*, it is in fact a labor supply to the *requester* that they are tracing out, as the MTurk worker has the full list of alternative MTurk jobs to choose from.

Experiment 1 (**E1**)**:** The first experiment, conducted in 2016, extends the Horton, Rand, and Zeckhauser (2011) design to experimentally test for worker left-digit bias.[14] We randomize wages for a census image classification task to estimate discontinuous labor supply elasticities at \$0.10. We posted a total of 5,500 unique HITS on MTurk tasks for \$0.10 that include a brief survey and a screening task where respondents view a digital image of a historical slave census schedule from 1850 or 1860 and answer whether they see markings in the "fugitives" column (for details on the 1850 slave census, see Dittmar and Naidu 2016). The respondent is then offered a choice of completing an additional set of classification tasks for a specific wage. This is close to the maximum number of unique respondents obtainable on MTurk within a month-long experiment.

Experiment 2 (**E2**)**:** We conducted a second experiment in 2022 to measure left-digit bias at the more salient and larger spike at \$1.00. This experiment uses a "honeypot" design, where treated workers see the job among the jobs available and a survey-taking task centered around 1.00. We first prescreened MTurk workers so as

[14] Preregistered as AEA RCT ID AEARCTR-0001349.

to exclude bots and then randomized the wage offered to each one for a job tailored to appear only in that worker's choice set of HITs while browsing MTurk. We posted 2,826 HITs this way, where the task was taking a survey experiment on hypothetical job preferences over 30 pairs of jobs with random characteristics (Dube et al. 2020) along with writing essays. This survey was designed to take enough time so as to induce variation in take-up around 1.00 but had a smaller overall take-up rate.

We pool data from both experiments and present results for the sample of sophisticated users (with more than 10 hours and primarily motivated by money, as defined by Horton, Rand, and Zeckhauser 2011) in order to increase the power to detect a significant labor-supply elasticity. A large sample of respondents who are more likely to be responsive to wage changes increases the precision with which we can test our hypothesis that wages differentially respond to one-cent increases at \$0.09 or \$0.99. We also show results including amateurs and disaggregating by experiment.

B. *Specification*

We begin by estimating experiment-specific regressions as follows:

$$(9) \qquad Accept_i = \beta_{0A} + \eta_0 w_i + \gamma_0\big(\mathbf{1}\{w_i \geq w_0\}_i\big) + \beta_1 T_i + \epsilon_i.$$

This corresponds to the specification from equation (3), which is linear in the perceived wage. Recall that an estimate of θ can be calculated as $\gamma_0/(\gamma_0 + w_0\eta_0)$. As discussed in Section II, we also want to estimate a constant elasticity specification, which is given by:

$$(10) \qquad Accept_i = \beta_{0A} + \eta_1 \log(w_i) + \gamma_1\big(\mathbf{1}\{w_i \geq w_0\}_i\big) + \beta_1 T_i + \epsilon_i.$$

In both cases, $w_0 \in \{10, 100\}$ denotes the threshold depending on the experiment. T_i is within-experiment controls (6 or 12 images for the 2018 experiment, wave for the 2022 experiment). We also compute estimates of θ as $\gamma_1/(\gamma_1 + \eta_1)$.

Next, we pool our experiments, estimating a regression with a single log wage coefficient to gain precision, but otherwise allow separate coefficients on the jumps across the two experiments, resulting in our preferred specification:

$$(11) \qquad Accept_i = \beta_{0A} + \eta_1 \log(w_i) + \gamma_1^{10}\big(\mathbf{1}\{w_i \geq 10\}_i \times \mathbf{1}\{E1\}_i\big)$$
$$+ \gamma_1^{100}\big(\mathbf{1}\{w_i \geq 100\}_i \times \mathbf{1}\{E2\}_i\big) + \beta_1 T_i$$
$$+ \boldsymbol{\beta}_2'\mathbf{X}_i + \boldsymbol{\beta}_3'\mathbf{X}_i \times \mathbf{1}\{E2\}_i + \epsilon_i.$$

In one variant, we additionally include worker-level controls, $\mathbf{X}_i$, interacted with the experiment indicators, where experiments are denoted by $E1$ and $E2$. We estimate θ_{10} and θ_{100} analogously. When estimating this specification, we additionally calculate a precision-weighted average of the two and report this as a pooled θ.

While specifications restricted to the sophisticated agents are our preferred ones, we also show specifications including the amateurs who use MTurk less than ten hours a week or whose primary motivation is not money. We can also include the data from Horton, Rand, and Zeckhauser (2011) to improve the precision of the η_1 estimate, further refining our power to detect left-digit bias.

Finally, we can also pool fully across the experiments estimating a single γ_1 coefficient along with a single semielasticity η_1, giving us the following specification:

$$(12)\ Accept_i = \beta_{0A} + \eta_1 \log(w_i)$$
$$+ \gamma_1\left(\mathbf{1}\{w_i \geq 10\}_i \times \mathbf{1}\{2018\}_i + \mathbf{1}\{w_i \geq 100\}_i \times \mathbf{1}\{2022\}_i\right)$$
$$+ \beta_1 T_i + \boldsymbol{\beta}_2' \mathbf{X}_i + \boldsymbol{\beta}_3' \mathbf{X}_i \times \mathbf{1}\{2022\}_i + \epsilon_i.$$

This has the cost of imposing stronger assumptions with the benefit of greater precision. As a practical matter, as we shall see, the point estimates of γ_1^{10} and γ_1^{100} are numerically close, so the primary effect is on precision. Based on this specification, we calculate a fully pooled θ coefficient as $\gamma_1/(\gamma_1 + \eta_1)$. Implicitly, this compares the average response to a one-cent increase at each threshold to the sample-averaged estimate of the semielasticity of labor.

C. *Results*

We begin by presenting the raw data for sophisticated respondents in the scatterplots in Figure 7, which shows the basic pattern. For both experiments, there is a clear positive slope. However, there are no discernible discontinuities at either \$0.10 or \$1.00.

Table 2 and Table 3 below show the key experimental results. The former shows experiment-specific estimates based on equation (10), with the wage variable in levels (columns 1and 2) or log (columns 3 and 4). The wage coefficients are positive and statistically significant in all samples and specifications. The implied labor supply elasticities are between 0.14 and 0.21. At the same time, these elasticities are small in magnitude indicating a high degree of monopsony power, consistent with the MTurk evidence in Dube et al. (2020).

The second set of coefficients (labeled as "Jump") estimate γ_1. Contrary to the left-digit bias hypothesis, the estimates in all four columns are negative but not statistically different from zero. We also report estimates of θ as described above. All the point estimates of θ are negative in sign and not distinguishable from zero. For E1, the log wage estimate rules out θ greater than 0.11 at the 5 percent level, while the level wage estimate rules out θ larger than 0.24. E2 rules out θ larger than 0.35 in logs or 0.36 in levels. (Note that while the coefficients on log wage are broadly similar across experiments (0.18 and 0.11), the level wage coefficients are of different orders of magnitude, which highlights why pooling across the two experiments is more feasible with the log wage specification.)

Next, to gain precision, column 1 in Table 3 jointly combines data from both experiments and estimates a common log wage coefficient using 11, while allowing each experiment to have a different jump at the relevant thresholds. The coefficient on the log wage estimate of 0.14 is more precise, with a smaller standard error than either individual experiment; at the same time, both jump coefficients continue to be small and statistically insignificant. Estimates here rule out θ_{10} greater than 0.21

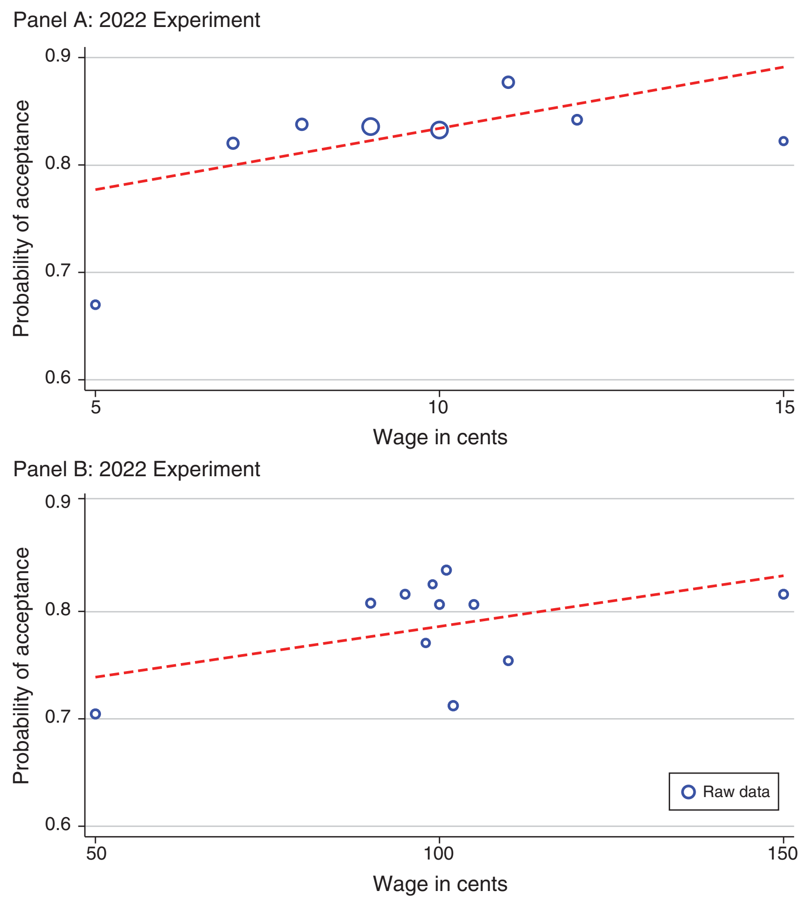

Figure 7. Distribution of Randomized Rewards in the MTurk Experiment and Resulting Probability of Task Acceptance

Notes: The figure shows the percentage change in probability of accepting the bonus task as a function of the wage for both experiments with the 2018 experiment on top and the 2022 experiment below. Slope of the fitted line is estimated from the pooled data, with a jump at the discontinuity, with no controls. Dots are scaled proportionally to the number of observations.

and θ_{100} greater than 0.22. A more precise labor supply elasticity estimate increases our power to bound the extent of possible left-digit bias. In addition, we report a precision-weighted average θ across the two estimates: The confidence interval for the pooled θ rule out values larger than 0.10.

Columns 2–5 show robustness of the estimates to inclusion of amateurs as well as adding data from the Horton, Rand, and Zeckhauser (2011) experiment. Estimates are mostly unchanged, though, as expected, inclusion of amateurs reduces precision of θ because amateur labor supply is less elastic. At the same time, use of all three experiments and all workers ($N = 8{,}430$) rules out a pooled θ greater than 0.14.

Columns 6 and 7 present estimates from a specification with a single jump and a single log wage coefficient based on equation (12). Note that the jump coefficients from the two experiments are not very different numerically, which suggest the main impact is on precision. Column 10 reports a θ of -0.22, ruling out values larger

TABLE 2—TASK ACCEPTANCE PROBABILITY BY OFFERED TASK REWARD ON MTURK: INDIVIDUAL EXPERIMENTS

	(1)	(2)	(3)	(4)
Log wage	0.178	0.111		
	(0.0633)	(0.0557)		
Wage			0.0147	0.00113
			(0.00688)	(0.000619)
Jump at \$0.10	−0.0321		−0.0174	
	(0.0270)		(0.0272)	
Jump at \$1.00		−0.0237		−0.0214
		(0.0263)		(0.0266)
DV mean	0.828	0.786	0.828	0.786
$\theta(10)$	−0.220		−0.135	
SE	(0.168)		(0.191)	
$\theta(100)$		−0.273		−0.232
SE		(0.325)		(0.298)
Observations	1,702	1,363	1,702	1,363
Spec.	E1	E2	E1-levels	E2-levels
Sample	Soph.	Soph.	Soph.	Soph.
Controls	N	N	N	N

Notes: E1 indicates the 2018 experiment testing the discontinuity at \$0.10, while E2 indicates the 2022 experiment testing the discontinuity at \$1.00. For columns 1 and 2, the wage variable is in logs, while for columns 3 and 4 it is in levels. The sample of sophisticates is restricted to Turkers who respond that they work more than 10 hours a week and that their primary motivation is money. Robust standard errors in parentheses.

TABLE 3—TASK ACCEPTANCE PROBABILITY BY OFFERED TASK REWARD ON MTURK-POOLED EXPERIMENTS

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Log wage	0.140	0.132	0.100	0.0703	0.108	0.141	0.108
	(0.0419)	(0.0411)	(0.0246)	(0.0273)	(0.0151)	(0.0417)	(0.0151)
Jump at \$0.10	−0.0190	−0.0143	−0.00526	−0.0102	−0.0231		
	(0.0225)	(0.0225)	(0.0198)	(0.0145)	(0.0124)		
Jump at \$1.00	−0.0321	−0.0300	−0.0207	0.0163	0.00562		
	(0.0246)	(0.0236)	(0.0230)	(0.0175)	(0.0164)		
Jump (pooled)						−0.0252	−0.0128
						(0.0186)	(0.0104)
DV mean	0.809	0.811	0.804	0.777	0.774	0.809	0.774
θ	−0.204	−0.182	−0.112	0.0462	−0.0777	−0.217	−0.135
SE	(0.155)	(0.159)	(0.185)	(0.171)	(0.110)	(0.156)	(0.116)
$\theta(10)$	−0.157	−0.122	−0.0555	−0.170	−0.271		
SE	(0.188)	(0.194)	(0.215)	(0.242)	(0.171)		
$\theta(100)$	−0.298	−0.295	−0.260	0.188	0.0493		
SE	(0.266)	(0.268)	(0.351)	(0.196)	(0.139)		
Observations	3,065	3,018	3,226	8,010	8,430	3,065	8,430
Spec.	E1/E2	E1/E2	E1/E2/HRZ	E1/E2	E1/E2/HRZ	E1/E2	E1/E2/HRZ
Sample	Soph.	Soph.	Soph.	All	All	Soph.	All
Controls	N	Y	N	N	N	N	N

Notes: E1 indicates the 2018 experiment testing the discontinuity at \$0.10, while E2 indicates the 2022 experiment testing the discontinuity at \$1.00. HRZ indicates the MTurk experiment conducted by Horton, Rand, and Zeckhauser (2011). The sample of sophisticates is restricted to Turkers who respond that they work more than 10 hours a week and that their primary motivation is money. Pooled estimates of θ are precision-weighted averages of jump-specific θ, except in columns 6 and 7 where a common jump coefficient is imposed. Robust standard errors in parentheses.

than 0.09. Finally, column 10 repeats this but with the sample expanded maximally (including amateurs and including data from Horton, Rand, and Zeckhauser 2011). The estimated θ is -0.14 and highly precise, again ruling out values greater than 0.09. These results are summarized graphically in Figure 9; we return to this figure below when we contextualize these estimates in relation to the extant literature.

We show additional specifications from the preanalysis plan in Supplemental Appendix F. Our preanalysis plan also included a test for whether task quality could exhibit left-digit bias, and we report those results in the Supplemental Appendix; we do not find any evidence of a discontinuity in task quality at \$0.10 either.

We interpret the evidence as strongly pointing away from a sizable left-digit bias on the workers' side. At the same time, we find a considerable amount of wage-setting power in this online labor market: Labor is fairly inelastically supplied to online employers, with an estimated elasticity η generally between 0.1 and 0.2.

D. *Panel Evidence from Oregon Administrative Data*

Using matched employer-employee data from Oregon, we estimate the responsiveness of employee separations to hourly wages and test for a discontinuity in separations at \$10.00/hour wage. The separation elasticity is widely used in the monopsony literature to measure the responsiveness of the supply of labor to a firm to the firm's posted wage (see Manning 2011).

We use a matched design that compares separation responses of new hires based on their starting wage, conditioning on a rich set of controls. For the 2003–2007 period, we start with the universe of all newly hired workers who have at least one quarter of tenure with their current employer (whom we refer to as "full quarter hires").[15] With the hiring date in quarters denoted as t, our sample includes workers who were still employed at the beginning of quarter $t + 1$. To focus on workers who were earning around \$10.00, we restrict our sample to those whose initial (full quarter) hourly wage at date $t + 1$ was between \$9.50 and \$10.59. We then calculate their four-quarter separation, $S_{i,t+4}$, which equals one if the worker leaves the firm by quarter $t + 4$. Our analysis examines how this separation probability varies with the starting full-quarter wage $w_{i,t+1}$ for workers hired with an initial wage during the same period and with similar characteristics at their previous job (at time $t - 1$).

A key concern in estimating this elasticity using observational data is the possibility that workers earning different wages may differ in other important dimensions. In our baseline specification, the vector $\mathbf{X}_{it}$ includes "workers controls:" an indicator for calendar quarter, the log of hourly wage (in \$0.10 intervals), and the log of weekly hours in their last full quarter at their previous job. In the most saturated specifications we also control for the value of the workers' (and previous employers') AKM effects.

Another concern arises from firm heterogeneity. As we saw in Section ID, firms engaging in greater round number bunching tend to be quite different. In the baseline specifications, "firm controls" include one-digit NAICS sectors and quintiles of the share of jobs in the range at their current employer that pay exactly \$10.00 in

[15] This restriction excludes workers with very marginal attachment to the workforce, such as seasonal workers. Notably, the spike at \$10.00 persists in this sample.

order to control for firm heterogeneity. This is particularly important because some firms (about a third in our sample) have no jobs paying exactly \$10.00, and these firms are quite different than those bunching at \$10.00, including ones that have all jobs in this range bunched at \$10.00. In addition, in some specifications we trim the sample to exclude firms where a very high share (more than 95 percent) or very low share (less than 5 percent) of wages are bunched at \$10.00 to ensure that the "common support" assumption holds. In our preferred specifications, we account for firm heterogeneity by interacting the quarter of hire with firm fixed effects.

Our most saturated specification considers workers who were hired within the same firm in the same quarter and who had previously earned very similar wages and worked similar hours at similar firms but are now earning somewhat different wages at their current jobs, and it considers their subsequent four-quarter separation probability. This matched panel design is similar to the approach developed by Bassier, Dube, and Naidu (2022).[16]

We estimate analogues of our experimental specifications above, regressing separations $S_{i,t+4}$ on log wages, and an indicator for earning more than \$10.00.

$$(13)\quad S_{i,t+4} = \beta_0 + \eta_1 \log(w_{i,t+1}) + \gamma_{1A}\mathbf{1}\{w_{i,t+1} \geq 10.00\}_i + \mathbf{\Lambda}'\mathbf{X}_{i,t+1} + \epsilon_{i,t+1}.$$

This specification tests for a jump in the labor supply at \$10 but constrains the slope to be the same on both sides, so left-digit bias is rejected if $\gamma_{1A} = 0$. We also calculate the left-digit bias parameter θ, which is approximaely equal to $\gamma_{1A}/(\gamma_{1A} + \eta_1)$.

The columns in Table 4 report the estimates from specifications with increasingly saturated controls. Using the dynamic monopsony steady-state assumption, where firm labor supply elasticity is twice the negative of the separation elasticity, the labor supply elasticities, η, range between 1.7 and 2.6 and all are statistically significant at the 5 percent level. These indicate a fair degree of monopsony power in the low-wage labor market in Oregon over this period and are consistent with findings in Bassier, Dube, and Naidu (2022).

For columns 2 and later, when we add an indicator for \$10/hour or more as a regressor, we do not see any indication that the separation rate changes discontinuously at \$10 in any specification, whether the sample includes all firms or is trimmed to exclude extremely bunched or extremely unbunched firms, whether we control for firm fixed effects, and whether we control for previous-job AKM values and values of worker fixed effects. In all cases, the jump coefficients are small, "wrong signed," and statistically indistinguishable from zero. We calculate the estimates for θ for each of these specifications. In the specifications with firm fixed effect (columns 4, 5, and 6), the 95 percent confidence intervals rule out estimates of θ greater than

[16]Bassier, Dube, and Naidu (2022) also examine the separation rates of newly hired workers as a function of their starting wage. One key difference is that they focus solely on wage policy differences across firms. Specifically, they regress the separation rate on the starting wage at time t, instrumented by the difference in average wages between the old and new firms, to isolate the firm-specific wage component. Their estimates of the labor supply elasticity for the 2003–2007 period, using a broader set of Oregon workers, is approximately 3.6 but with smaller estimates in the low-wage sector, which aligns with our findings. However, our main interest lies in detecting left-digit bias by testing for discontinuities in the worker-level separation elasticity around \$10.00, which requires using individual-level, rather than solely firm-level, wage variation. Moreover the bias from heterogeneity of firms paying exactly a round number is not a concern in their case, but it is for us here.

Table 4—Oregon Matched Panel Estimates: Separation Response to Starting Wage around $10.00/Hour Jobs

	(1)	(2)	(3)	(4)	(5)	(6)
Log wage	−0.381	−0.455	−0.615	−0.564	−0.425	−0.426
	(0.087)	(0.208)	(0.293)	(0.170)	(0.171)	(0.179)
Jump at 10.00		0.009	0.021	0.015	0.016	0.014
		(0.014)	(0.020)	(0.010)	(0.011)	(0.011)
θ		−0.248	−0.518	−0.375	−0.580	−0.484
		(0.359)	(0.405)	(0.226)	(0.342)	(0.334)
η	1.650	1.971	2.606	2.440	1.838	1.797
	(0.374)	(0.897)	(1.225)	(0.736)	(0.738)	(0.755)
Observations	107,631	107,623	71,362	63,785	63,765	53,163
Worker controls		Y	Y	Y	Y	Y
Firm controls		Y	Y			
Trimmed sample (round share)			Y			Y
Firm fixed effects				Y	Y	Y
Lag WFE FFE value					Y	Y

Notes: Sample includes all "full quarter hires," with one quarter tenure at their new job who were employed at a different company two quarters prior, whose hourly wages fall between $9.50 and $10.59, and whose first quarter hours exceeded the twenty-fifth percentile (184 hours). The outcome is four-quarter separation which takes on one if the person leaves their job within four quarters of the start date. The mean four-quarter separation rate in this sample is 0.49. Standard controls include quarter fixed effects, log hours and log wage at the last job two quarters prior, quintiles in the share of jobs bunched at exactly $10.00 at the current employer, and a one-digit NAICS industry dummy. Firm fixed effects indicates a fixed effect for the current employer interacted with quarter dummies. The trimmed sample excludes workers at firms where very few (less than 5 percent) or very high (greater than 95 percent) of jobs in this range are bunched at $10.00. η is the elasticity of labor supply facing the firm. θ is the inattention parameter capturing left-digit bias. Robust standard errors are clustered at the one cent wage bin level.

0.17 in all cases. Finally, in Figure 8, we plot the coefficient on each $0.10 dummy between $9.50 and $10.40 associated with equation (13), with firm fixed effects and the full sample (i.e., column 4) of Table 4. Consistent with the findings in the table, there is a general negative relationship between wages and separation, with a similar implied separations elasticity as the baseline specification above. However, there is no indication of a discontinuous drop in the separation rate going from $9.90 to $10.00. Similar to the experimental evidence, there seems to be no discontinuity in labor supply at $10.00. And most estimates rule out a high degree of left-digit bias.

While our primary analysis focuses on starting wages for new hires, we also exploit variation in wages that comes from subsequent raises. For instance, consider two individuals who both were hired at the same firm at date t earning the same wage under $10, but between $t+1$ and $t+2$ one person got a raise that placed her above $10, while the other's raise was not sufficient to cross that threshold. We could ask what that does to the separation probability between dates $t+2$ and $t+3$.

In Supplemental Appendix Table A.1, column 1, we begin by reproducing our starting wage analysis for comparability focusing on those with one quarter of tenure, using firm-by-quarter fixed effects. Then, in columns 2 and 3, we restrict to those with three quarters of tenure and further control for their log wage, and an indicator for the wage exceeding $10.00 when they were at a tenure of one quarter. Now the tenure of two quarter log wage and jump at $10.00 is identified only from the raise that occurs between the first and second quarter at the firm. Column 3 further controls for previous job-based characteristics. The θ in both specifications are

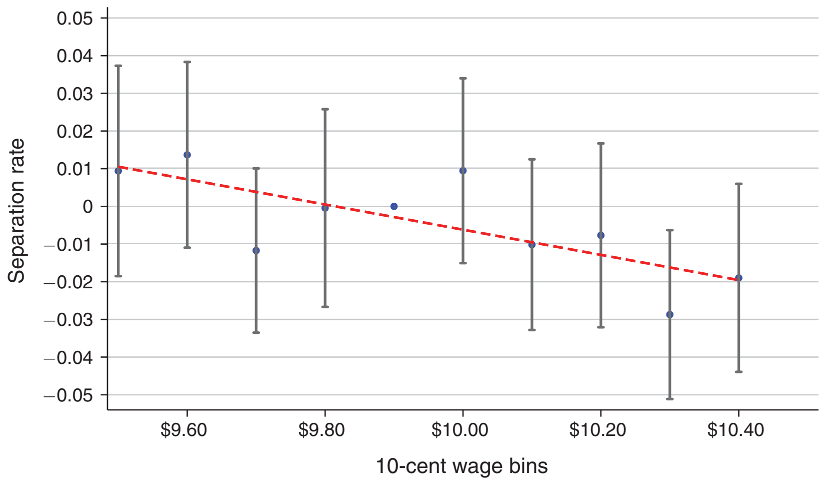

Figure 8. Oregon Separation Response by Wage Bin

Notes: Estimated from quarterly matched worker-establishment data from Oregon 2003–2007. Sample includes all "full quarter hires"—that is, with one quarter tenure at their new job—who were employed at a different company two quarters prior, whose hourly wages fall between \$9.50 and \$10.59, and whose first quarter hours exceeded the twenty-fifth percentile (184 hours). The outcome is four-quarter separation which takes on one if the person leaves their job within four quarters of start date. Four-quarter separation rate with blue circles are coefficient on \$0.10 wage bin dummies relative to \$9.90 (the bin right below \$10.00) and control for fully saturated interactions of quarter-of-hire and firm fixed effects, as well as log hours and log wage at the last job two quarters prior. Robust standard errors are clustered at the one-cent wage bin level.

negative in sign, statistically indistinguishable from zero, and rule out values larger than 0.13 at the 95 percent level. Workers who started with similar wages under \$10 were no less likely to quit when raises placed them above the \$10 threshold.

E. *Comparison with Product Market Estimates*

A natural question is whether our experiments allow us to reject levels of left-digit bias estimated in the product market literature. As discussed above, Strulov-Shlain (2023) parameterizes left-digit bias in demand with θ, the inattention parameter. He specifies the empirical demand function to be $\log(Q+1) = \beta^0 p + \beta^1 \lfloor p \rfloor$, and recovers $\theta = \beta^1/(\beta^1 + \beta_0)$.

Overall, Strulov-Shlain (2023) finds a θ of 0.22 among customers at a supermarket in his data, indicating a substantial amount of left-digit bias. (Interestingly, he finds that supermarkets do not sufficiently bunch prices at \$0.99 given the extent of consumer left-digit bias.) List et al. (2023), estimating similar specifications for Lyft riders with experimental variation, finds a lowest θ of 0.44. A nonproduct market comparison comes from Lacetera, Pope, and Sydnor (2012), who evaluate change in used car prices around 10,000 thresholds in odometer readings; they find an overall estimate of θ equal to 0.31.

Figure 9 presents comparable estimates of θ from our analysis using both the experimental variation and observational estimates. The figure draws from a select set of experimental estimates presented in Tables 2 and 3.

The individual experiments rule out a high degree of left-digit bias and in all cases rule out a θ as large as 0.36; in some cases they rule out any θ larger than 0.11.

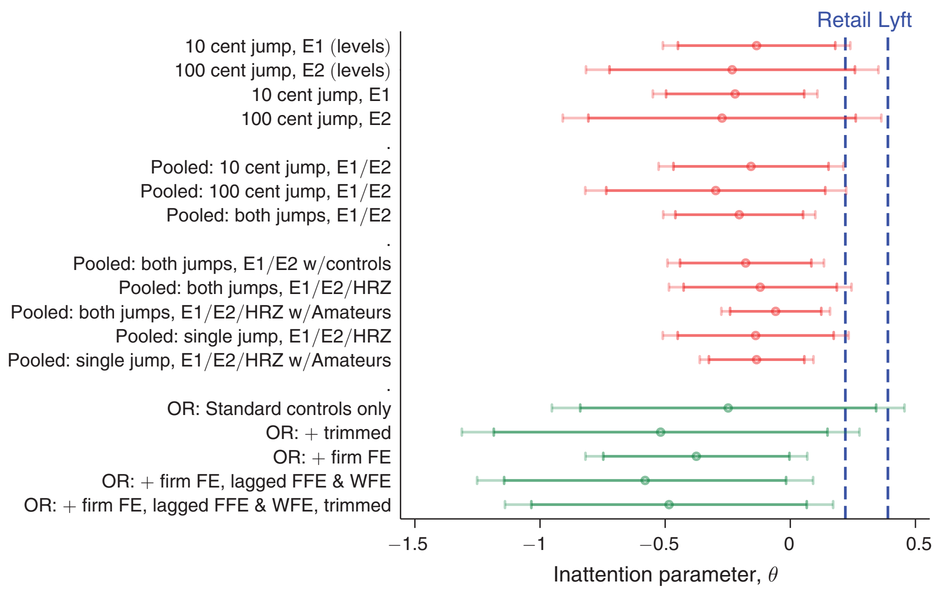

Figure 9. Comparison with Product Market Inattention Parameters

Notes: Shows estimates and confidence intervals (95 percent and 90 percent) of the attention parameter θ. The red points correspond to specifications estimated on the MTurk experiments from Table 2. E1 indicates the 2018 experiment testing the discontinuity at \$0.10, while E2 indicates the 2022 experiment testing the discontinuity at \$1.00. HRZ denotes that data from the MTurk experiment conducted in Horton, Rand, and Zeckhauser (2011) is included. Unless "Amateurs" is specified, all specifications are estimated on the sample of "sophisticates": Turkers who respond that they work more than 10 hours a week and that their primary motivation is money. The "pooled, both jumps" specifications estimate separate jump coefficients at \$0.10 and \$1.00 discontinuities and then take a precision-weighted average of the two estimates of θ. The "pooled, single jump" specifications impose a single coefficient on both discontinuities and calculate θ under the assumption of constant η and constant γ across both experiments. The green points show estimates from the administrative matched worker-firm data in Oregon under progressively finer specifications as in Table 4. The vertical lines correspond to estimates of the inattention parameter from retail in Strulov-Shlain (2023) and from Lyft in List et al. (2023).

Pooling across experiments greatly increases our power to more sharply bound the extent of left-digit bias through a more precise estimate of the labor supply elasticity. Pooling across the E1 and E2 and estimating a common log wage coefficient, the 95 percent confidence intervals rule out θ_{10} above 0.21 and θ_{100} above 0.22—right around the retail estimate. The confidence intervals for the precision-weighted pooled θ rule out anything greater than 0.10, well under both product market comparisons (Strulov-Shlain 2023; List et al. 2023).

As the figure shows, most pooled estimates across a range of specifications and samples rule out the key product market estimates of θ. The most precise estimate comes from pooling all data, including from Horton, Rand, and Zeckhauser (2011), while estimating a single jump parameter and a single log wage coefficient and ruling out any θ greater than 0.09. Overall the experimental estimates are inconsistent with left-digit bias of a substantial magnitude.

Turning to separation response estimates from the Oregon data, our preferred specifications (with firm FE controls) rule out θ larger than 0.17 at the 95 percent confidence level. The least saturated specifications and no trimming on share bunched are less precise (and also less reliable for the reasons explained above), but they do rule out θ greater than 0.34 at the 90 percent confidence level.

On the whole, the weight of evidence from both the experimental and observational data do not detect comparable levels of left-digit bias found in the product market. An interesting direction for future research is exploring why left-digit bias is so much larger in product prices than hourly wages; we conjecture it is because any given product is a small share of expenditure while hourly wages are a large share of income for most wage workers, and so rational inattention could account for the differences.[17] In contrast, in the context of a job, the stakes are larger for workers—making mistakes more costly. For example, Caldwell, Haegele, and Heining (2025) show that when individual-level bargaining is feasible, workers often engage in many bargaining rounds before accepting or rejecting an offer, even when the wage gains resulting in such bargaining are relatively small. This is consistent with a high-stakes environment where behavioral biases on the worker side may be uncommon, whether wages are set by bargaining or wage posting.

IV. Incorporating Employer Optimization Frictions

Given the lack of evidence for worker left-digit bias, we now consider the other possible explanation for bunching: employer misoptimization. We extend the model to allow employers to "benefit" by paying a round number despite lowered profits.[18] While consistent with employers preferring to pay round numbers, it could reflect internal fairness constraints or administrative costs internal to the firm. These could be transaction costs involved in dealing with round numbers, managers' cognitive costs, or administrative costs facing a bureaucracy.

The heterogeneity in firm bunching presented so far already points toward some misoptimization being important. Section ID showed that bunching was more prevalent in firms less likely to survive as well as experiencing slower employer growth and higher turnover, just as Reyes (2024) showed in Brazil. Further, we showed that firms in industries with lower quality management were more likely to have jobs bunched at \$10. All of these suggest that bunching reflects a constraint on firm maximization with the resulting prediction that firm performance ought to be lower when there is a high degree of bunching.

We parameterize employer misoptimization by δ, measuring the willingness to forgo profit in order to pay a round number. The presence of δ modifies the profit function to be

$$\pi(w,p) = (p - w) l(w,p) e^{\delta \mathbf{1}_{\{w=w_0\}}}, \tag{14}$$

where δ is the percentage "gain" in profits from paying the round number.[19] This specification parallels that in Chetty (2012), who restricts optimization frictions to be constant fractions of optimal consumer expenditure (in the nominal model), except when applied to the employer's choice of wage for a job rather than a

[17] Matějka and McKay (2015) provide foundations for discrete choice that incorporate inattention; see Gabaix (2019) for applications of inattention to a wide variety of behavioral phenomena, including left-digit bias.

[18] It would be equivalent to assuming that firms suffer an effective loss from not paying a round number.

[19] While we do not microfound why employers may have preferences for paying a particular round number, this may reflect inattention among wage setters. For example, Matějka (2015) shows that rationally inattentive monopoly sellers will choose a discrete number of prices even when the profit-maximizing price is continuous.

consumer's choice of a consumption-leisure bundle. In the taxable income model, optimization frictions parameterize the lack of responsiveness to tax incentives, while in our model they parameterize the willingness to forgo profits in order to pay a round number.

Given (5) and (14), profits from paying a wage w to a worker with marginal product p can be written as

$$\begin{aligned} \pi(w,p) &= (p - w)\frac{w^{\eta}}{C}e^{\gamma \mathbf{1}_{\{w \geq w_0\}}}e^{\delta \mathbf{1}_{\{w=w_0\}}}k(p) \\ &= (p - w)\,l^*(w,p)\,e^{\gamma \mathbf{1}_{\{w \geq w_0\}}}e^{\delta \mathbf{1}_{w=w_0}}, \end{aligned} \tag{15}$$

which can be written in terms of the nominal model ρ as

$$\delta + \gamma \mathbf{1}_{\{w^*(p) < w_0\}} > \ln\rho(w^*(p),p) - \ln\rho(w_0,p). \tag{16}$$

This shows that bunching is more likely the greater the left-digit bias of workers and the optimization cost for employers. The optimization bias is symmetric whether the latent wage is above or below the round number, but left-digit bias is asymmetric because it only has an impact if the latent wage is below the round number. The right-hand side of (16) can be approximated using the following second-order Taylor series expansion of $\rho(w_0,p)$ about $w^*(p)$:[20]

$$\begin{aligned} \ln\rho(w_0,p) \simeq{}& \ln\rho(w^*,p) + \frac{\partial \ln\rho(w^*,p)}{\partial w}[w_0 - w^*] \\ &+ \frac{1}{2}\frac{\partial^2 \ln\rho(w^*,p)}{\partial w^2}[w_0 - w^*]^2. \end{aligned} \tag{17}$$

The first-order term is zero by the definition of the latent wage and the envelope theorem (Akerlof and Yellen (1985) use this idea to explain price and wage rigidity). Using (6) the second-order term can be written as

$$\frac{\partial^2 \ln\rho(w^*,p)}{\partial w^2} = -\frac{\eta(1+\eta)}{w^{*2}}. \tag{18}$$

Substituting (18) into (17) and then into (16) leads to the following expression for whether a firm pays the round number:

$$\frac{1}{2}\left[\frac{w_0 - w^*}{w^*}\right]^2 \equiv \frac{\omega^2}{2} \leq \frac{\delta + \gamma \mathbf{1}_{\{w^* < w_0\}}}{\eta(1+\eta)}. \tag{19}$$

The left-hand side of (19) implies that the size of the loss in nominal profits from bunching is increasing in the square of the proportional distance of the latent wage from the round number (ω). The right-hand side tells us that, for a given latent

[20] One can use the actual profit function instead of the approximation, but the difference is small for the parameters we use, and the approximation has a clearer intuition.

wage, whether a firm will bunch depends on the extent of left-digit bias as measured by γ (only relevant for wages below the round number), the extent of optimization frictions as measured by δ, and the degree of competition in the labor market as measured by η. The extent of labor market competition matters because the loss in profits from a suboptimal wage are greater the more competitive the labor market.

In Supplemental Appendix D we formally show that, with $\gamma = 0$, the missing mass is distributed symmetrically from above and below w_0, which is consistent with our empirical bunching estimates as well as our experimental evidence on left-digit bias. Taking $\gamma = 0$ as given, our estimates of the proportion of firms who bunch for each latent wage identify the CDF of $z_1 = \delta/\left[\eta(1+\eta)\right]$, but do not allow us to identify the distributions of δ and η separately since that requires additional assumptions about their joint distribution. This simplest place to start is to assume a single value of η and a two-part distribution of δ equal to δ^* for bunchers and 0 for others. This yields an expression for the $\delta^* - \eta$ locus for a given extent of bunching as measured by ω:

$$\frac{\omega^2}{2} = \frac{\delta^*}{\eta(1+\eta)}. \tag{20}$$

Figure 10 plots the δ^*, η locus as expressed in equation (20) for the sample width of the normalized bunching interval ω based on the pooled administrative data as well as for the 90 percent confidence interval around it. We can see visually that as we consider higher values of δ^* the range of admissible ηs increases and becomes larger in value. However, even for sizable δ^*s, the implied values of the labor supply elasticity are often modest, implying at least a moderate amount of monopsony power.

A basic consistency check is whether plausible magnitudes of δ^*, within "economic standard errors" (Chetty 2012) of perfect profit maximization, imply estimates of η that are within the range of existing monopsony estimates. We calculate estimates of η assuming δ^* equal to 0.01, 0.05 or 0.1 in rows A, B, and C of Supplemental Appendix Table C.2, respectively. The implied labor supply elasticity η varies between 1.3 and 5.1 when we vary δ^* between 0.01 and 0.1. These are well within the range of recent estimates of η for the low-wage US labor market from the literature (Sokolova and Sorensen 2021; Bassier, Dube, and Naidu 2022).

We also have direct estimates of η from our experiments and administrative data. η in the matched worker-firm Oregon data is in the 2–3 range and implies a δ of between 0.02 and 0.04. The amount of money being lost by firms mispricing labor seems similar or somewhat smaller than other estimates from the recent behavioral industrial organization literature (DellaVigna and Gentzkow (2019) and Strulov-Shlain (2023) find 10.2–11.7 percent and 1–4 percent of profits are lost due to employer mispricing in retail, respectively). Overall, the profit losses implied by the optimization friction-based explanation for round number bunching are consistent with the evidence from other types of firm misoptimization.

Returning to the heterogeneity in Section ID, one might wonder whether the variation across types of firms is explained more by η or by δ. As an example, we can use our preferred specification above (with firm fixed effects) to estimate separation response and η by management score tercile, and we can confirm that there is little heterogeneity in the degree of monopsony power across industries by this variable: η is around 0.970 for the bottom quartile versus 0.995 for the top quartile.

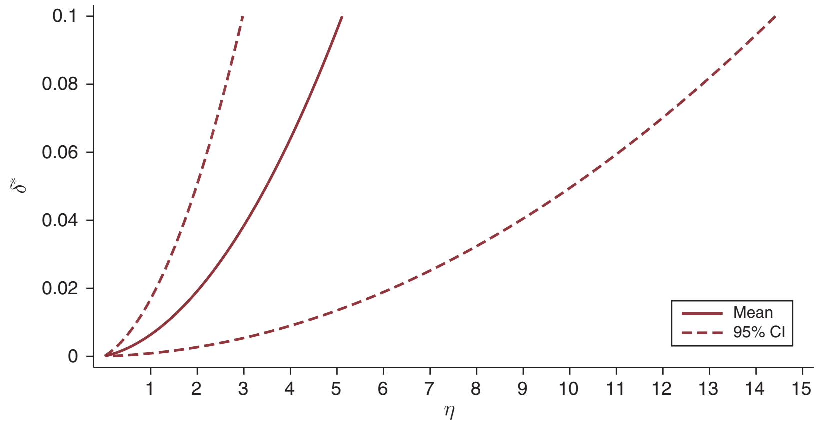

Figure 10. Relationship between Labor Supply Elasticity (η) and Optimization Frictions (δ), Consistent with the Empirical Extent Size of Bunching (ω): Administrative Hourly Wage Data from Minnesota, Oregon, and Washington

Notes: The solid, red, upward-sloping line shows the locus of the labor supply elasticity η and optimization frictions $\delta^* = E[\delta | \delta > 0]$ consistent with the extent of bunching ω estimated using the administrative hourly wage data from Minnesota, Oregon, and Washington between 2003:I and 2007:IV, as described in equation (5) in the paper. The dashed lines are the 95 percent confidence intervals estimated using 500 bootstrap replicates.

Similarly the much higher bunching among small firms cannot be explained by differences in the elasticity of labor supply faced by employers. For instance, while the degree of bunching among firms with 50 or fewer workers is substantially higher than in the full sample, the separation elasticities are quite similar. For firms with 50 or fewer workers, the separation elasticity using our baseline specification is -2.047 (SE $=$ 0.198), while for the full sample it is -2.163 (SE $=$ 0.203).

At the same time, the Figure 6 also provides an indication that bunching is more prevalent among firms with high AKM firm effects than for firms with low firm effects. In dynamic monopsony models like Burdett and Mortensen (1998), labor supply elasticity is likely to be lowest at firms near the top of the (residual) wage distribution, reflecting the relative scarcity of better-paying competitors that can poach workers. We empirically confirm that separation elasticities are smaller in magnitude for firms in the top quintile of AKM firm effects, as compared to the bottom quintile.[21] Better-paying firms facing limited competition bear fewer consequences for deviating from their profit-maximizing wage.

MTurk data show a stark version of the trade-off. In Supplemental Appendix Figure C.4 we plot the excess and missing mass from MTurk. Compared to the low labor supply elasticity recovered from our experiments above, the ω implied by this figure (0.18) can be rationalized by a very small δ (0.003). The substantial monopsony power employers have on MTurk means that only a small degree of misoptimization is required to rationalize the large degree of observed bunching.

In Supplemental Appendix D, we additionally show how flexible distributions of η, γ, and δ can be recovered from the amount and symmetry of the missing mass

[21] While the separation elasticity using our baseline specification is 1.214 (SE $=$ 0.436) for the top quintile, it is 2.212 (SE $=$ 0.286) for the bottom quintile.

around a round number. These results suggest that in contexts where only granular histograms of wages are available (e.g., restricted-use administrative data or historical payroll records), researchers may be willing to make assumptions about firm misoptimization in order to recover bounds on the degree of employer market power.

In Supplemental Appendix Table C.4 we explore robustness to different assumptions on heterogeneity in δ and η, including allowing for arbitrary heterogeneity in δ or heterogeneity in η but not δ, and finally a semiparametric approach where we impose a lognormal distribution on δ but an arbitrary nonparametric distribution on η. We also examine robustness of the estimates to alternative specifications of the latent distribution of wages in Supplemental Appendix Table C.5. This includes controlling for secondary modes at \$0.50 and \$0.25 bins and varying the polynomial used to estimate the latent distribution.

V. Conclusion

Significantly more US workers are paid exactly round numbers than would be predicted by a smooth distribution of marginal productivity and the neoclassical model. We document this fact in administrative data, mitigating any issues due to measurement error, and it is present even in Amazon MTurk, an online spot labor market where there are no regulatory constraints or long-term contracts.

Unlike in the product market, where numerous researchers have rationalized prices ending in \$0.99 with "left-digit bias," we rule out worker inattention with two high-powered experiments on MTurk as well as observational data on separation elasticities from Oregon. We instead find evidence that this is consistent with employer misoptimization: Bunching firms are smaller, less likely to survive, and have worse management. We also quantify the combinations of monopsony power and employer misoptimization necessary to rationalize any degree of bunching: Only moderate amounts of misoptimization are required to explain observed bunching given estimates of monopsony.

Finally, as we show in Supplemental Appendix G, round number bunching of wages is relevant for a number of labor market policies. In monopsonistic settings, wage inertia means that minimum-wage hikes create larger spillovers when they cross a round-number threshold, and that small payroll-tax increases are typically absorbed by firms. Our estimates suggest that many firms are leaving money on the table. At the same time, pervasive monopsony supports "satisficing" wage-setting practices, where employers settle on familiar pay benchmarks instead of fine-tuning each wage. In this way, our findings echo the institutionalist view that due to impediments to competitive forces, wages can remain indeterminate within a range (Lester 1952).

REFERENCES

Abowd, John M., Francis Kramarz, and David N. Margolis. 1999. "High Wage Workers and High Wage Firms." *Econometrica* 67 (2): 251–333.

Akerlof, George A., and Janet L. Yellen. 1985. "A Near-Rational Model of the Business Cycle, with Wage and Price Inertia." *Quarterly Journal of Economics* 100 (S): 823–38.

Allen, Eric J., Patricia M. Dechow, Devin G. Pope, and George Wu. 2016. "Reference-Dependent Preferences: Evidence from Marathon Runners." *Management Science* 63 (6): 1657–72.

Anderson, Eric T., and Duncan I. Simester. 2003. "Effects of \$9 Price Endings on Retail Sales: Evidence from Field Experiments." *Quantitative Marketing and Economics* 1 (1): 93–110.

Azar, José, Ioana Marinescu, and Marshall Steinbaum. 2022. "Labor Market Concentration." *Journal of Human Resources* 57 (S): S167–S199.
Babcock, Linda, William J. Congdon, Lawrence F. Katz, and Sendhil Mullainathan. 2012. "Notes on Behavioral Economics and Labor Market Policy." *IZA Journal of Labor Policy* 1 (1): 2.
Backus, Matthew, Thomas Blake, and Steven Tadelis. 2019. "On the Empirical Content of Cheap-Talk Signaling: An Application to Bargaining." *Journal of Political Economy* 127 (4): 1599–628.
Bassier, Ihsaan, Arindrajit Dube, and Suresh Naidu. 2022. "Monopsony in Movers: The Elasticity of Labor Supply to Firm Wage Policies." *Journal of Human Resources* 57 (S): S50–S86.
Basu, Kaushik. 1997. "Why Are So Many Goods Priced to End in Nine? And Why This Practice Hurts the Producers." *Economics Letters* 54 (1): 41–44.
Basu, Kaushik. 2006. "Consumer Cognition and Pricing in the Nines in Oligopolistic Markets." *Journal of Economics and Management Strategy* 15 (1): 125–41.
Berger, David, Kyle Herkenhoff, and Simon Mongey. 2022. "Labor Market Power." *American Economic Review* 112 (4): 1147–93.
Blomquist, Sören, Whitney K. Newey, Anil Kumar, and Che-Yuan Liang. 2021. "On Bunching and Identification of the Taxable Income Elasticity." *Journal of Political Economy* 129 (8): 2320–43.
Bloom, Nicholas, and John Van Reenen. 2007. "Measuring and Explaining Management Practices across Firms and Countries." *Quarterly Journal of Economics* 122 (4): 1351–408.
Burdett, Kenneth, and Dale T. Mortensen. 1998. "Wage Differentials, Employer Size, and Unemployment." *International Economic Review* 39 (2): 257–73.
Caldwell, Sydnee, Ingrid Haegele, and Jörg Heining. 2025. "Bargaining and Inequality in the Labor Market." NBER Working Paper 33396.
Caldwell, Sydnee, and Emily Oehlsen. 2018. "Monopsony and the Gender Wage Gap: Experimental Evidence from the Gig Economy." Unpublished.
Chen, Daniel L., and John J. Horton. 2016. "Are Online Labor Markets Spot Markets for Tasks? A Field Experiment on the Behavioral Response to Wage Cuts." *Information Systems Research* 27 (2): 403–23.
Chetty, Raj. 2012. "Bounds on Elasticities with Optimization Frictions: A Synthesis of Micro and Macro Evidence on Labor Supply." *Econometrica* 80 (3): 969–1018.
Cho, David. 2018. "The Labor Market Effects of Demand Shocks: Firm-Level Evidence from the Recovery Act." Unpublished.
Cho, Sungjin, and John Rust. 2010. "The Flat Rental Puzzle." *Review of Economic Studies* 77 (2): 560–94.
DellaVigna, Stefano, and Matthew Gentzkow. 2019. "Uniform Pricing in US Retail Chains." *Quarterly Journal of Economics* 134 (4): 2011–84.
DellaVigna, Stefano, and Devin Pope. 2018."What Motivates Effort? Evidence and Expert Forecasts." *Review of Economic Studies* 85 (2): 1029–69.
Derenoncourt, Ellora, and David Weil. 2025. "Voluntary Minimum Wages: The Local Labor Market Effects of National Retailer Policies." *Quarterly Journal of Economics*. https://doi.org/10.1093/qje/qjaf016.
Dittmar, Jeremiah, and Suresh Naidu. 2016. "Contested Property: Fugitive Slaves in the Antebellum US South." Unpublished.
Dube, Arindrajit, Jeff Jacobs, Suresh Naidu, and Siddharth Suri. 2020. "Monopsony in Online Labor Markets." *American Economic Review: Insights* 2 (1): 33–46.
Dube, Arindrajit, Alan Manning, and Suresh Naidu. 2017. *Left-Digit Bias in Wage Perceptions and Labor Supply in an Online Labor Market*. AEA RCT Registry. https://doi.org/10.1257/rct.1349-2.0
Dube, Arindrajit, Alan Manning, and Suresh Naidu. 2025. *Data and Code for: "Monopsony and Employer Misoptimization Explain Why Wages Bunch at Round Numbers."* American Economic Association; distributed by Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/E226928V1.
Gabaix, Xavier. 2019. "Behavioral Inattention." In *Handbook of Behavioral Economics: Applications and Foundations*, Vol. 2, edited by B. Douglas Bernheim, Stefano DellaVigna, and David Laibson, 261–343. Elsevier.
Gabaix, Xavier, and David Laibson. 2006. "Shrouded Attributes, Consumer Myopia, and Information Suppression in Competitive Markets." *Quarterly Journal of Economics* 121 (2): 505–40.
Goldfarb, Avi, and Mo Xiao. 2011. "Who Thinks about the Competition? Managerial Ability and Strategic Entry in US Local Telephone Markets." *American Economic Review* 101 (7): 3130–61.
Hazell, Jonathon, Christina Patterson, Heather Sarsons, and Bledi Taska. 2022. "National Wage Setting." NBER Working Paper 30623.
Heidhues, Paul, and Botond Kőszegi. 2008. "Competition and Price Variation when Consumers Are Loss Averse." *American Economic Review* 98 (4): 1245–68.

Heidhues, Paul, and Botond Kőszegi. 2018. "Behavioral Industrial Organization." In *Handbook of Behavioral Economics: Applications and Foundations* 1, Vol. 1, edited by B. Douglas Bernheim, Stefano DellaVigna, and David Laibson, 517–612. Elsevier.

Hortaçsu, Ali, and Steven L. Puller. 2008. "Understanding Strategic Bidding in Multi-Unit Auctions: A Case Study of the Texas Electricity Spot Market." *RAND Journal of Economics* 39 (1): 86–114.

Horton, John J., David G. Rand, and Richard J. Zeckhauser. 2011. "The Online Laboratory: Conducting Experiments in a Real Labor Market." *Experimental Economics* 14 (3): 399–425.

Jones, Edward D. 1896. "Round Numbers in Wages and Prices." *Publications of the American Statistical Association* 5 (35–36): 111–41.

Kingsley, Sara Constance, Mary L. Gray, and Siddharth Suri. 2015. "Accounting for Market Frictions and Power Asymmetries in Online Labor Markets." *Policy and Internet* 7 (4): 383–400.

Kleven, Henrik Jacobsen. 2016. "Bunching." *Annual Review of Economics* 8: 435–64.

Kleven, Henrik J., and Mazhar Waseem. 2013. "Using Notches to Uncover Optimization Frictions and Structural Elasticities: Theory and Evidence from Pakistan." *Quarterly Journal of Economics* 128 (2): 669–723.

Kline, Patrick, Neviana Petkova, Heidi Williams, and Owen Zidar. 2019. "Who Profits from Patents? Rent-Sharing at Innovative Firms." *Quarterly Journal of Economics* 134 (3): 1343–404.

Kuziemko, Ilyana, Michael I. Norton, Emmanuel Saez, and Stefanie Stantcheva. 2015. "How Elastic Are Preferences for Redistribution? Evidence from Randomized Survey Experiments." *American Economic Review* 105 (4): 1478–508.

Lacetera, Nicola, Devin G. Pope, and Justin R. Sydnor. 2012. "Heuristic Thinking and Limited Attention in the Car Market." *American Economic Review* 102 (5): 2206–36.

Lamadon, Thibaut, Magne Mogstad, and Bradley Setzler. 2022. "Imperfect Competition, Compensating Differentials, and Rent Sharing in the US Labor Market." *American Economic Review* 112 (1): 169–212.

Lester, Richard A. 1946. "Wage Diversity and Its Theoretical Implications." *Review of Economics and Statistics* 28 (3): 152–59.

Levy, Daniel, Dongwon Lee, Haipeng Allan Chen, Robert J. Kauffman, and Mark Bergen. 2011. "Price Points and Price Rigidity." *Review of Economics and Statistics* 93 (4): 1417–31.

List, John A., Ian Muir, Devin Pope, and Gregory Sun. 2023. "Left-Digit Bias at Lyft." *Review of Economic Studies* 90 (6): 3186–237.

Lester, Richard A. 1952. "A Range Theory of Wage Differentials." *ILR Review* 5(4): 483–500.

Manning, Alan. 2011. "Imperfect Competition in the Labor Market." In *Handbook of Labor Economic*, Vol. 4B, edited by David Card and Orley Ashenfelter, 973–1041. Elsevier

Manning, Kenneth C., and David E. Sprott. 2009. "Price Endings, Left-Digit Effects, and Choice." *Journal of Consumer Research* 36 (2): 328–35.

Matějka, Filip. 2015. "Rationally Inattentive Seller: Sales and Discrete Pricing." *Review of Economic Studies* 83 (3): 1125–55.

Matějka, Filip, and Alisdair McKay. 2015. "Rational Inattention to Discrete Choices: A New Foundation for the Multinomial Logit Model." *American Economic Review* 105 (1): 272–98.

Pope, Devin G., Jaren C. Pope, and Justin R. Sydnor. 2015. "Focal Points and Bargaining in Housing Markets." *Games and Economic Behavior* 93: 89–107.

Reyes, Germán. 2024. "Coarse Wage-Setting and Behavioral Firms." *Review of Economics and Statistics*. https://doi.org/10.1162/rest_a_01470.

Romer, David. 2006. "Do Firms Maximize? Evidence from Professional Football." *Journal of Political Economy* 114 (2): 340–65.

Shaw, Aaron D., John J. Horton, and Daniel L. Chen. 2011. "Designing Incentives for Inexpert Human Raters." In *Proceedings of the ACM 2011 Conference on Computer Supported Cooperative Work*, edited by Pamela Hinds, John C. Tang, Jian Wang, Jakob Bardram, and Nicolas Ducheneaut, 275–84. Association for Computing Machinery.

Snir, Avichai, Daniel Levy, Alex Gotler, and Haipeng Allan Chen. 2012. "Not All Price Endings Are Created Equal: Price Points and Asymmetric Price Rigidity." Preprint, SSRN. http://dx.doi.org/10.2139/ssrn.2167662.

Sokolova, Anna, and Todd Sorensen. 2021. "Monopsony in Labor Markets: A Meta-Analysis." *ILR Review* 74 (1): 27–55.

Strulov-Shlain, Avner. 2023. "More than a Penny's Worth: Left-Digit Bias and Firm Pricing." *Review of Economic Studies* 90 (5): 2612–45.

Thomas, Manoj, and Vicki Morwitz. 2005. "Penny Wise and Pound Foolish: The Left-Digit Effect in Price Cognition." *Journal of Consumer Research* 32 (1): 54–64.

Zwick, Eric. 2021. "The Costs of Corporate Tax Complexity." *American Economic Journal: Economic Policy* 13 (2): 467–500.